

Engineering Economics MEE2003 — MTE Study Pack

Built 2026-09-25 from what you uploaded: 10 decks (L1–L19), the L13 EOQ practice sheet, the handwritten L14–19 notes and the course hand-out. No assignment has been uploaded yet, so the **professor's own problems** play the assignment's role. Every section opens with one of them, you guess first, then the section teaches what the question needs.

Accuracy. You asked for a pack, not research, so this is built from the slides. The content is standard engineering economics (Panneerselvam, the prescribed text, agrees). **Every number in this pack was recomputed by script.** Where a slide's arithmetic is wrong, the pack gives the correct figure and marks it **⚠ slide error**. There are 3 of them: 04 \$2, 08 \$5, 09 \$4.

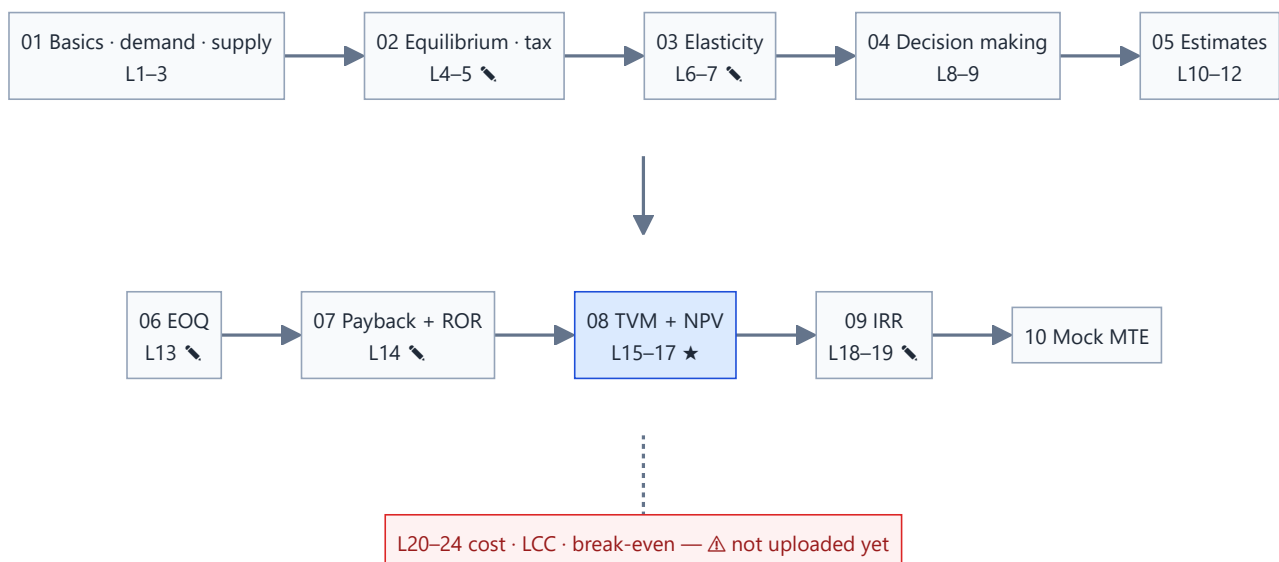
What's on the MTE

Source	Says
Hand-out lecture plan	MID TERM EXAM printed after L24
Marks	MTE 30 · CWS 30 (quiz, assignment) · ETE 40
Question format	unknown. No PYQ yet. The ECE2108 MTE the same week was A: 2+2+3 · B: 3 × 5 · C: 1 × 8 , and 10 uses that

MTE syllabus = L1–L24. This pack covers **L1–L19** (everything uploaded).

⚠ Not covered, because no deck has arrived: L20 cost concepts and elements of cost · L21 life cycle cost and project financing · L22–24 **break-even analysis (3 lectures of numericals)**. Upload those decks and they get added. Break-even alone is likely worth a full long question.

The path



Priority if time is short: 08 → 06 → 07 → 09 → 02 → 03 (all numerical, which is where the marks are) → 05 → 04 → 01. The last three are theory and can be answered by writing the definitions and points.

Calculator: every numerical here needs powers such as 1.1^5 . Practise $(1 + i)^n$ and the annuity factor $(1 - (1+i)^{-n}) / i$ on your own calculator before the exam.

Every professor-set question → where it is answered

Source	Question (short)	File
L1	Case: solar farm / high-speed rail, economics vs engineering economics view	01
L2	Technical vs economic efficiency	01
L4-5 P1	$X_d = \frac{1}{2}(5-P)$, $X_s = 2P-3$; tax ₹6/5	02
L4-5 P2	$P = 5-2X$, $P = \frac{1}{2}(X+5)$; subsidy ₹5/2	02
L4-5 P3	$Q_d = 200-10P$, $Q_s = 50+15P \rightarrow Q_s' = 100+15P$	02
L4-5 P4	$3Q+4P = 24$, $P = \frac{1}{4}Q+3$; tax ₹1/3	02
L7 P1	Price ₹50→48, demand 100→110: E_p	03
L7 P2	Income ₹100→200, qty 25→30: E_y	03
L7 P3	Advertising ₹6000→12000, qty 8000→10000: E_a	03
L8-9	Cloud server: on-demand vs reserved (EAC) ⚠ slide error	04
L8-9	Petrol car vs electric car, 8-year total cost	04
L8-9	Audi vs BMW: rational decision steps	04
L10-12	ROM: office building, software duration, gadget, web app	05
L10-12	Budget estimate: building, software product	05
L10-12	Parametric: house, steel shafts, PCB assembly	05
L10-12	Analogous: CMS 10,000 → 15,000 lines	05
L10-12	Bottom-up: CMS task list + 20% contingency	05
L13	Car dealer: $D = 5000$, $S = 15000$, $H = 500$	06
L13	16000 units, ₹2, $S = ₹45$, carrying 10%: EOQ + cycle time	06
L13 sheet Q1	Batteries: current cost, EOQ, orders/yr	06
L13 sheet Q2	Indian Telecom: EOQ, orders, reorder point	06
L13 sheet Q3	20 units/yr, lead time 3 months: EOQ, ROP	06
L13 sheet Q4	Bicycle tyres: EOQ, orders, total cost	06
L14	Payback: $\$10500 / \$2500 \cdot \text{beverage equipment} \cdot \$200,000 \text{ uneven} \cdot ₹50000 \text{ uneven}$	07
L14	ROR of the \$10500 project · lighting retrofit (notes)	07
L15	₹75,000 today vs ₹1,00,000 in 5 years · \$1000 in 5 years at 10%	08
L16-17	\$2000 project, $\$100 \times 3 + \2500	08

Source	Question (short)	File
L16-17	Project A vs B, \$10,000, 10% ⚠ slide error	08
L16-17	Standard vs high-efficiency furnace (10% and 30%)	08
L16-17	Project A vs B by payback AND NPV (–1200 ...)	08
L16-17	Real estate ₹200000, ₹30000/yr, 12%	08
Notes	₹150000, ₹50000 × 5, 10%	08
L18-19	Machine –\$50,000, \$15,000 × 5 at 10/15/20% + IRR	09
L18-19	Interpolation demo: +200 at 10%, –100 at 15%	09
L18-19	Equipment ₹500000, ₹160000 × 4, salvage ₹50000 ⚠ slide error	09

How to use each file

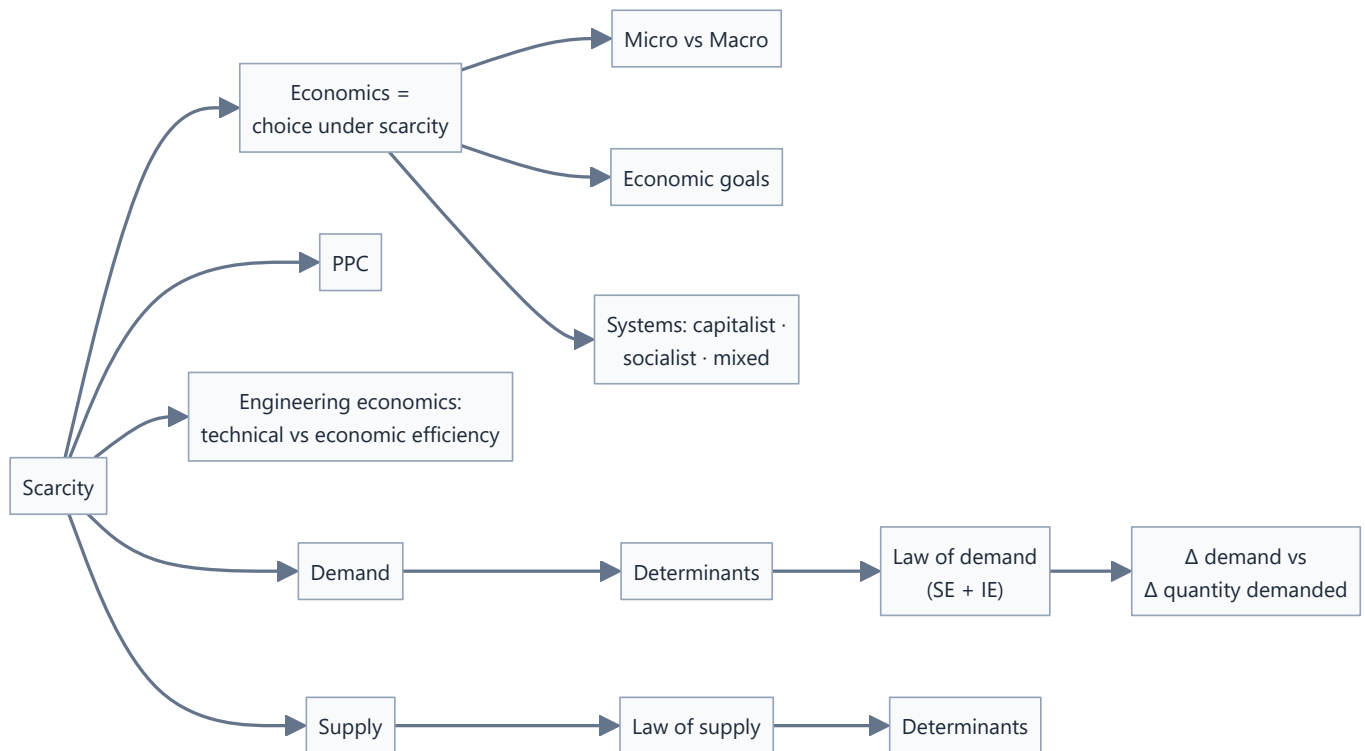
When you hit a **Q ▶**, write a one-line guess or first step on paper **before** reading on. A wrong guess still makes the answer stick better than reading it cold. Every file ends with a **Self-test**, and its answers are at the very bottom. Whatever you miss there is what you revise last.

How numericals are marked: (1) the formula written out, (2) the substitution shown, (3) the answer with units, (4) a **one-line decision** (accept/reject, which option). Leaving out the decision line is the easiest way to lose a mark.

01 — Economics basics, demand and supply (L1–L3)

Theory file. On the exam this shows up as 2–3 mark "define / differentiate / explain with a diagram" questions. The diagrams (demand curve, supply curve, PPC) are worth drawing every time.

Map



1. Scarcity and what economics is

Q ▶ (L1) Wants are unlimited. Resources are limited. What problem does that create, and what do people have to do about it?

Scarcity: wants exceed the resources available to satisfy them. Faced with scarcity, people must **choose**. So:

- **Economics** is the study of the choices people make to cope with scarcity, or equivalently the allocation of scarce resources among alternative uses to satisfy human wants.
- The **basic economic problems** are **resource allocation** (what to produce, and whether labour-intensive or capital-intensive) and the **distribution of income**.

	Microeconomics	Macroeconomics
Studies	individual units: consumers, firms, workers, investors, and their markets	aggregates: national output and its growth, interest rates, unemployment, inflation
Example question	how a firm prices one product	why inflation rose this year

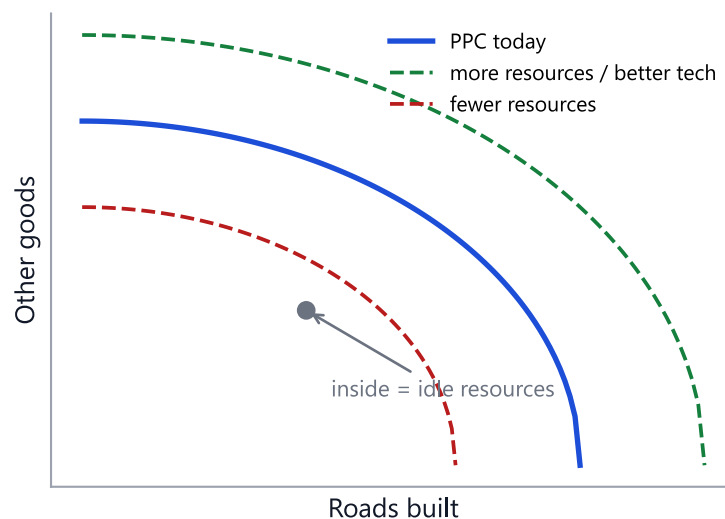
Economic goals: high employment · price stability · equitable income distribution · growth. They are **not always complementary**. Cutting unemployment sharply tends to raise inflation.

2. Production possibilities curve (PPC)

Q ▶ A country can build roads or produce something else with fixed resources. What does the curve of all the maximum combinations look like, and what makes it move?

The **PPC** plots the maximum output combinations of two goods with the given resources and technology. It is bowed outward. The economy is on the curve when it is efficient and inside it when resources are idle.

Event	PPC
More resources, or better technology	shifts outward (growth)
Fewer resources, or degraded technology / low investment	shifts inward (low growth)



PPC: growth shifts it out, lost resources shift it in, a point inside means idle resources.

3. Economic systems

An **economic system** is the set of arrangements and institutions that meets a society's two objectives: **increasing the availability of resources** and **ensuring their economic use**.

	Capitalist	Socialist	Mixed
Decides what/how to produce	free market (demand and supply)	a regulatory authority (e.g. a planning commission). Private property and inheritance are abolished	both
Merits	self-regulating, faster growth, operative efficiency, flexible	social security, fewer fluctuations, coordinated development, fewer social disputes	aims to get the benefits of both

	Capitalist	Socialist	Mixed
Demerits	income/wealth inequality, unequal opportunity, distorted production, merit goods unprofitable so not produced, wasteful competition, loss of welfare values	weak economic incentives, slow growth, low labour productivity, low per-capita income (the slide also lists cyclical fluctuations)	its features have to be chosen by judging the market mechanism's effects

4. Engineering economics and the two efficiencies

Q ▶ (L2) A diesel engine's efficiency is below 100%. A business's efficiency must be **above** 100%. How can both be true?

They are two different efficiencies:

Technical efficiency = system output / system input (physical, e.g. energy)
ALWAYS < 100% (friction, incomplete combustion)

Economic efficiency = system worth / system cost
worth = annual revenue; cost = total annual expenses
MUST be > 100% for the business to survive and grow

Engineering economics covers the methods for taking economic decisions that **minimize cost and/or maximize benefit** to a business. It runs from recognising a need, through developing and evaluating alternatives, to deciding and executing efficiently.

Economics vs engineering economics (L1 case studies: 100 MW solar farm, high-speed rail).

- **Economics view:** the **whole society**, meaning jobs, emissions, income distribution, regional growth, externalities, and effects on prices.
- **Engineering economics view:** the **project's own financial feasibility**, meaning investment, operating cost, revenue, **NPV, IRR, payback**.
- Exam line: *"Economics gives the social picture; engineering economics decides whether the project pays. Good decisions use both."*

5. Demand

Q ▶ (L2) "Desire? Ability? Willingness?" You would like a sports car but can't pay for it. Is that demand?

No. That is only **desire**. The chain is:

- **Desire:** a wish to possess something.
- **Want:** desire backed by purchasing power and by willingness to buy.
- **Demand:** the quantity of a commodity a consumer wishes to buy **at a given price**, backed by purchasing power.

Three rules about a quantity demanded:

1. It is the **desired** purchase, not the actual one.
2. It is a **flow**, measured per period of time. "10 oranges" means nothing. "10 per week" does.
3. It has meaning only **at a given price**. "10 per week at ₹100/dozen" is a complete statement.

Static vs dynamic analysis

Static	Dynamic
variables not dated , "timeless"	variables dated , and time matters
one point of equilibrium	an economy in motion, with unexpected changes (new taxes, subsidies)
D_t and S_t both depend on today's price P_t	$D_t = f(P_t)$, $S_t = g(P_{t-1})$, $D_t = S_t$: today's supply depends on yesterday's price (a time lag)

Determinants of an individual's demand (the demand function $D_x = f(P_x, P_y, P_z, M, T)$):

Symbol	Factor	Effect
P_x	own price	$\uparrow$ price $\rightarrow$ $\downarrow$ quantity (law of demand)
M	income	normal good : $\uparrow M \rightarrow \uparrow Q$. Inferior good : $\uparrow M \rightarrow \downarrow Q$
P_y, P_z	prices of related goods	substitutes (used instead) and complements (used together)
T	taste	a taste for the good $\rightarrow$ more demand, and tastes can be seasonal

Market demand adds two more: **size of the population**, and **income distribution** (a richer top share means more demand for luxuries, a richer poor share means more demand for wage goods). Only one factor is changed at a time. The rest are held constant (*ceteris paribus*).

6. Law of demand and why the curve slopes down

Q ▶ (L3) Coke goes from ₹20 to ₹25 while Pepsi stays at ₹20. Pepsi's price didn't move. Why do people buy more Pepsi?

Because Pepsi became **relatively** cheaper. That is the **substitution effect**.

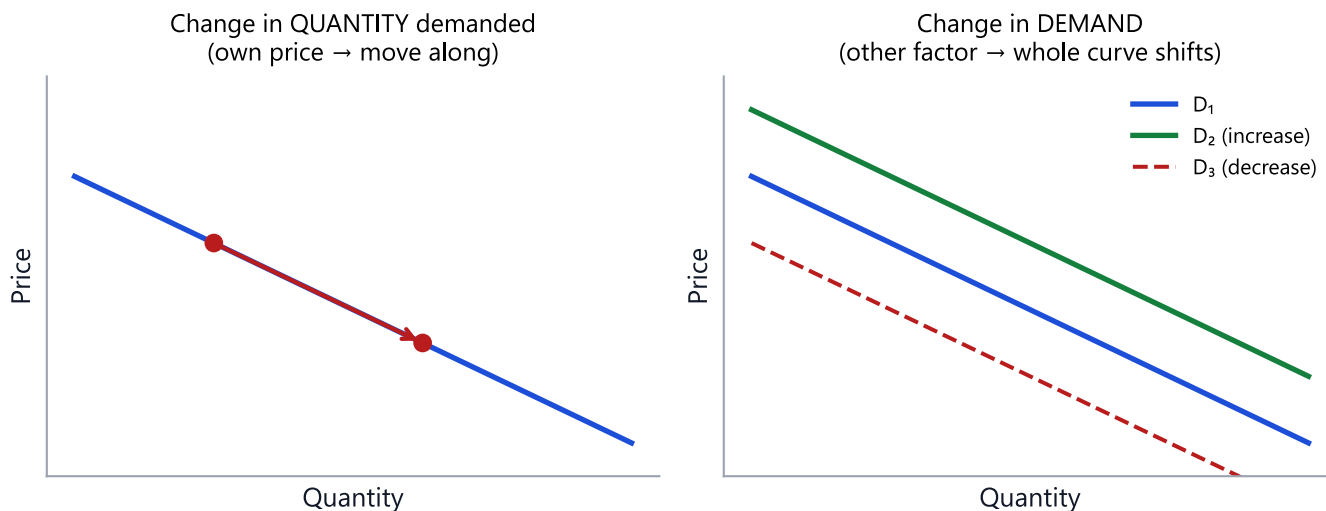
Law of demand: other things equal, quantity demanded varies **inversely** with price, so the demand curve **slopes downward to the right**.

It slopes down because of the **price effect = substitution effect + income effect (PE = SE + IE)**:

- **Substitution effect (SE)**: a price change alters **relative** prices, and buyers switch toward the relatively cheaper good.
- **Income effect (IE)**: a price fall raises **real income** (purchasing power), so the buyer can afford more.
- The two operate **simultaneously**.

Change in quantity demanded vs change in demand. This is a near-certain 2-marker:

	Change in quantity demanded	Change in demand
Cause	the good's own price	anything else (income, tastes, related prices, population)
On the graph	movement along the same curve	the whole curve shifts
Names	expansion (price $\downarrow$) / contraction (price $\uparrow$)	increase (shift right) / decrease (shift left)



Left: a movement along the curve (own price). Right: shifts of the whole curve (any other factor).

7. Supply

Supply is the quantity producers are **willing to sell** at each price, per unit of time.

Law of supply: price and quantity supplied move in the **same direction**, because a higher price means more profit (profit = TR – TC, with TR = p·q and TC = q·AC). Example (L3): pens at ₹2 → 25 thousand/month, up to ₹6 → 70 thousand.

Supply function: $Q_s = f(P_1, P_2 \dots P_n, F_1 \dots F_m, T, G)$

Determinant	Effect on supply of this good
own price P_1	$\uparrow \rightarrow \uparrow$ (the most immediate one)
prices of other goods $P_2 \dots P_n$	$\uparrow \rightarrow \downarrow$ (other goods become more attractive to produce)
prices of factors F (the cost of production)	$\uparrow \rightarrow \downarrow$
technology T	better → lower cost → $\uparrow$
producer's goal G	shapes how much is produced

Self-test

1. Give one example each of a normal good and an inferior good, and state the income effect on each.
2. A fall in the price of petrol raises car sales. Is that a movement along the car-demand curve or a shift? Which way?
3. Write the formulas for technical and economic efficiency, and say which must exceed 100%.
4. Write the dynamic demand–supply model and say what makes it dynamic.
5. Give two merits and two demerits of a capitalist economy.
6. What shifts a PPC inward?

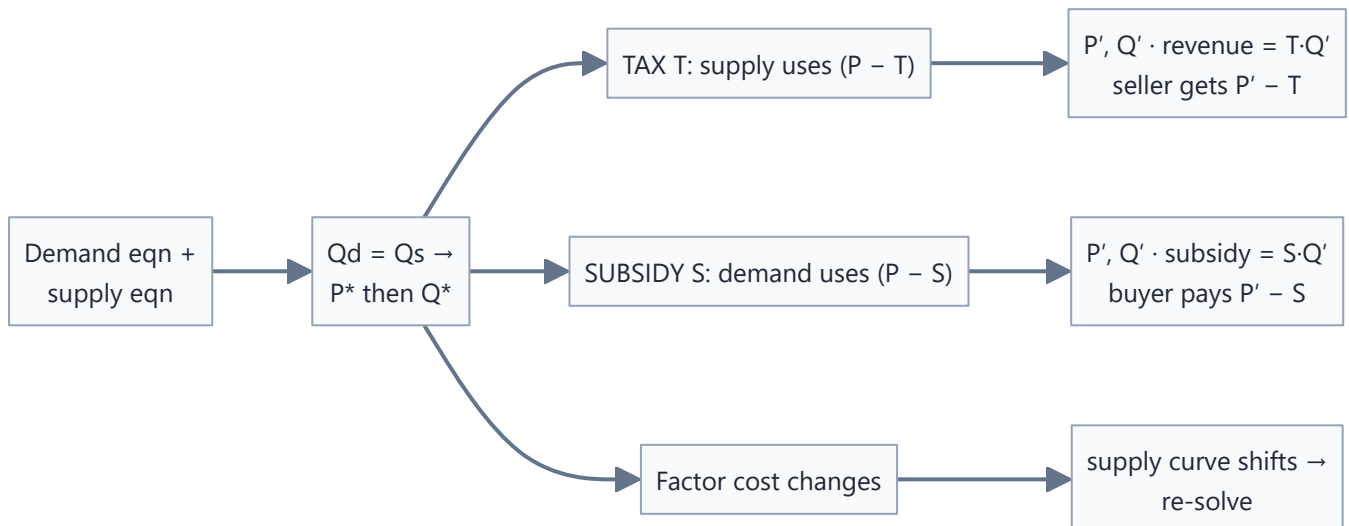
Answers

1. Normal good: branded clothing ($\uparrow \text{income} \rightarrow \uparrow Q$). Inferior good: coarse grain / cheap bus travel ($\uparrow \text{income} \rightarrow \downarrow Q$).
2. A **shift to the right** (an increase in demand). Petrol is a **complement** of cars, and its price is not the cars' own price.
3. Technical = output/input, always $< 100\%$. Economic = worth/cost, which **must be $> 100\%$** .
4. $D_t = f(P_t)$, $S_t = g(P_{t-1})$, $D_t = S_t$. The variables are dated and supply reacts to **last period's price** (a time lag).
5. Merits: self-regulating, faster growth (also operative efficiency, flexibility). Demerits: inequality of income/wealth, merit goods not produced (also wasteful competition, distorted production).
6. A decrease in available resources, degraded technology, or low investment.

02 — Market equilibrium, tax and subsidy (L4–L5)

Numerical file. The 4 deck problems are all answered here, and every answer was checked with exact fractions. These are pure algebra once you know the one rule for tax and the one rule for subsidy.

Map



1. Equilibrium

Equilibrium: the price at which **quantity demanded = quantity supplied**. Solve the two equations simultaneously. Always find **P first, then substitute to get Q**, and check it in **both** equations.

2. Tax: which curve moves and why

Q ▶ (L4-5) The market price is ₹100 and the tax is ₹10 per unit. How much does the producer actually get, and which equation changes?

₹100 – ₹10 = **₹90**. The producer responds to the price **net of tax**, so the **supply** equation changes:

$$S_t = g(P_t - T) \quad \text{replace every } P \text{ in the supply equation by } (P - T)$$

The market price rises by less than T, the quantity falls, the government collects **T × Q'**, and the seller **realises P' – T**.

3. Subsidy (the deck's convention)

A subsidy is Government assistance that lowers the buyer's effective price. The deck writes it into **demand**:

$$D_t = f(P_t - S) \quad \text{replace } P \text{ in the demand equation by } (P - S)$$

P' is then the **market price the seller receives**. The buyer effectively pays $P' - S$. The total subsidy is $S \times Q'$.

4. Problem 1 (tax) [L4-5 P1]

$X_d = \frac{1}{2}(5 - P)$, $X_s = 2P - 3$. (i) equilibrium price and quantity (ii) new price and quantity with a tax of ₹6/5 per unit (iii) total tax revenue (iv) price actually realised by the seller.

Q ▶ Set up (i) before reading on.

(i) $\frac{1}{2}(5 - P) = 2P - 3 \rightarrow 5 - P = 4P - 6 \rightarrow 5P = 11 \rightarrow P = 11/5 = ₹2.20$

$X_d = \frac{1}{2}(5 - 11/5) = \frac{1}{2} \cdot 14/5 = 7/5 = 1.4 \text{ units}$ (check: $X_s = 22/5 - 3 = 7/5 \checkmark$)

(ii) New supply: $X_s' = 2(P' - 6/5) - 3 = 2P' - 27/5$

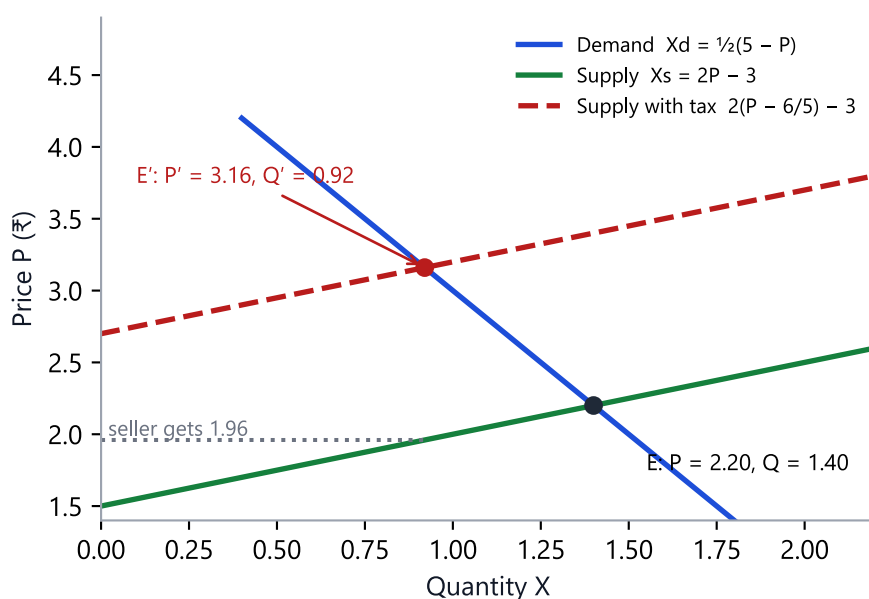
$\frac{1}{2}(5 - P') = 2P' - 27/5 \rightarrow 5 - P' = 4P' - 54/5 \rightarrow 5P' = 79/5 \rightarrow P' = 79/25 = ₹3.16$

$Q' = 2(79/25) - 27/5 = 158/25 - 135/25 = 23/25 = 0.92 \text{ units}$

(iii) Tax revenue = $T \times Q' = 6/5 \times 23/25 = 138/125 = ₹1.104$

(iv) Seller realises $P' - T = 79/25 - 30/25 = 49/25 = ₹1.96$

Read the result: the price rose from 2.20 to 3.16, which is **less than the ₹1.20 tax**. Buyers carry 0.96 of it and sellers 0.24 (2.20 $\rightarrow$ 1.96).



Problem 1 drawn to scale: the tax shifts supply up by ₹1.20, but the price rises only ₹0.96.

5. Problem 2 (subsidy) [L4-5 P2]

Demand $P = 5 - 2X$, supply $P = \frac{1}{2}(X + 5)$. (i) equilibrium (ii) new price and quantity with a subsidy of ₹5/2 per unit (iii) total subsidy.

Q ▶ Which equation gets the subsidy, and how?

(i) $5 - 2X = \frac{1}{2}(X + 5) \rightarrow 10 - 4X = X + 5 \rightarrow X = 1, P = 5 - 2 = ₹3$

(ii) Demand becomes $(P' - 5/2) = 5 - 2X' \rightarrow P' = 15/2 - 2X'$

$15/2 - 2X' = \frac{1}{2}(X' + 5) \rightarrow 15 - 4X' = X' + 5 \rightarrow X' = 2$

$P' = \frac{1}{2}(2 + 5) = 7/2 = ₹3.50$, the market price the seller receives. The buyer effectively pays $3.50 - 2.50 = ₹1$.

(iii) Total subsidy = $S \times X' = 5/2 \times 2 = ₹5$

6. Problem 3 (supply shift) [L4-5 P3]

$Q_d = 200 - 10P$, $Q_s = 50 + 15P$. (i) equilibrium. (ii) A factor-input price change gives $Q_s' = 100 + 15P'$. Analyse the new equilibrium against the old.

(i) $200 - 10P = 50 + 15P \rightarrow 150 = 25P \rightarrow P = ₹6$, $Q = 200 - 60 = 140$ units

(ii) $200 - 10P' = 100 + 15P' \rightarrow 100 = 25P' \rightarrow P' = ₹4$, $Q' = 200 - 40 = 160$ units

Analysis (write this sentence): the intercept rose from 50 to 100, so more is supplied at every price. The input got **cheaper** and **supply increased** (the curve shifted right). The **equilibrium price fell (6 → 4) and the quantity rose (140 → 160)**.

7. Problem 4 (tax) [L4-5 P4]

Demand $3Q + 4P = 24$, supply $P = \frac{1}{4}Q + 3$. (i) equilibrium (ii) tax ₹1/3 per unit (iii) tax revenue (iv) price realised by the seller.

Q ▶ Rewrite the demand as $P = \dots$ first.

Demand: $4P = 24 - 3Q \rightarrow P = 6 - \frac{3}{4}Q$

(i) $6 - \frac{3}{4}Q = \frac{1}{4}Q + 3 \rightarrow Q = 3, P = \frac{3}{4} + 3 = 15/4 = ₹3.75$

(ii) Supply with tax: $P' - \frac{1}{3} = \frac{1}{4}Q + 3 \rightarrow P' = \frac{1}{4}Q + 10/3$

$6 - \frac{3}{4}Q = \frac{1}{4}Q + 10/3 \rightarrow Q' = 6 - 10/3 = 8/3 \approx 2.67$ units

$P' = \frac{1}{4} \cdot 8/3 + 10/3 = 2/3 + 10/3 = ₹4$

(iii) Revenue = $\frac{1}{3} \times 8/3 = 8/9 \approx ₹0.89$

(iv) Seller realises $4 - \frac{1}{3} = 11/3 \approx ₹3.67$

Traps

- If the equation is in $P = \dots$ form (inverse supply), the tax goes in as $(P - T) = \dots$, i.e. $P' = \text{old RHS} + T$. If it is in $Q = \dots$ form, replace P by $(P - T)$. Both say the same thing.
- **Revenue uses the new quantity Q'** , never the old one.
- Keep fractions until the last line. The deck's answers are fractions (79/25, 138/125).
- Always state the **seller's price = $P' - T$** separately. It is its own sub-question in 2 of the 4 problems.

Self-test

1. $Q_d = 100 - 5P$, $Q_s = 20 + 3P$. Find the equilibrium.
 2. Same market, with a tax of ₹8 per unit. Find the new P' , Q' , the tax revenue and the seller's price.
 3. Same market (no tax), with a subsidy of ₹4 per unit on the demand side (deck convention). Find P' , Q' , what the buyer pays, and the total subsidy.
 4. In one line: why does the price rise by less than the tax?
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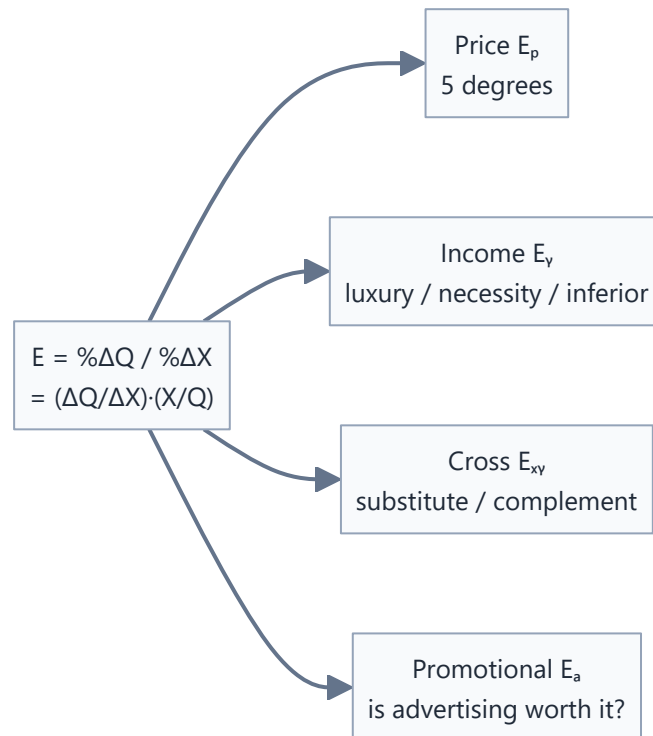
Answers

1. $100 - 5P = 20 + 3P \rightarrow \mathbf{P = 10, Q = 50}$.
2. $Q_s' = 20 + 3(P - 8) = 3P - 4 \rightarrow 100 - 5P = 3P - 4 \rightarrow \mathbf{P' = 13, Q' = 35}$. Revenue = $8 \times 35 = \mathbf{₹280}$. Seller gets $13 - 8 = \mathbf{₹5}$.
3. $Q_d' = 100 - 5(P - 4) = 120 - 5P = 20 + 3P \rightarrow \mathbf{P' = 12.5, Q' = 57.5}$. Buyer pays $12.5 - 4 = \mathbf{₹8.5}$. Subsidy = $4 \times 57.5 = \mathbf{₹230}$.
4. The quantity falls as the price rises, so the burden is **shared**: buyers pay part (a higher P') and sellers absorb part (a lower net price).

03 — Elasticity of demand (L6–L7)

All 3 deck numericals are answered here. Every elasticity uses the same formula shape. Learn it once and change the variable.

Map



1. The one formula

Q ▶ (L6) The price rises by 10%. We know demand falls, but by more than 10% or by less?

That depends on the good, and measuring it is exactly what **elasticity** does. It is the responsiveness of quantity demanded to a change in price, income or the price of a related good.

$$E = \frac{\% \text{ change in } Q}{\% \text{ change in } X} = \frac{\Delta Q / Q}{\Delta X / X} = \frac{\Delta Q}{\Delta X} \times \frac{X}{Q}$$

ΔQ = new - original quantity

Q = ORIGINAL quantity

ΔX = new - original X

X = ORIGINAL price / income / ad spend

Use the **ORIGINAL values** in X/Q . The deck does this in every problem.

2. The four elasticities

Elasticity	X	Reading the answer
Price E_p	own price P	normally negative. Judge its size
Income E_y	income Y (or M)	the sign and size classify the good
Cross E_{xy}	price of another good Y (quantity of X)	the sign classifies the relationship
Promotional E_a	advertising spend A	positive → the advertising expands demand

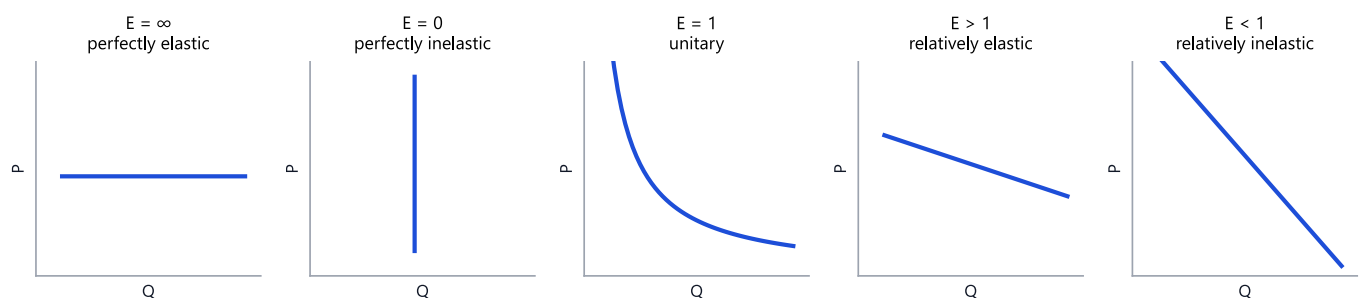
Degrees of price elasticity (draw these curves if asked):

	Value	Curve
Perfectly elastic	$E = \infty$	horizontal line
Perfectly inelastic	$E = 0$	vertical line
Unitary elastic	$E = 1$	rectangular hyperbola ($P \times Q$ constant)
Relatively elastic	$E > 1$	flatter
Relatively inelastic	$E < 1$	steeper

Income elasticity types (the deck's labels):

E_y	Good
> 1	luxury / superior
$= 1$	"semi-luxury" (deck's term. Standard name: unit income elastic)
$0 < E_y < 1$	necessity (a normal good)
$= 0$	income doesn't change demand (cheap necessities, e.g. salt)
< 0	inferior good

Cross elasticity: > 0 **substitutes** (tea and coffee: Y's price ↑ → X's demand ↑). < 0 **complements** (car and petrol: Y's price ↑ → X's demand ↓). $= 0$ **unrelated**. (The L6 slide explains the negative case using "income". The variable is the price of Y.)



The five degrees of price elasticity.

3. Price elasticity [L7 P1]

Price reduced from ₹50 to ₹48 per unit to attract more customers. Demand increased from 100 to 110 units. Calculate the price elasticity of demand.

Q ▶ Write ΔQ , ΔP and the originals before substituting.

$$\Delta Q = 110 - 100 = 10 \cdot \Delta P = 48 - 50 = -2 \cdot P = 50, Q = 100$$

$$E_p = (\Delta Q / \Delta P) \times (P / Q) = (10 / -2) \times (50 / 100) = -2.5$$

$|E_p| = 2.5 > 1 \rightarrow$ **relatively elastic**. A 4% price cut raised quantity by 10%. (For an elastic good, a price cut raises total revenue: $50 \times 100 = 5000 \rightarrow 48 \times 110 = 5280$.)

4. Income elasticity [L7 P2]

Income rises from ₹100 to ₹200. The quantity purchased rises from 25 to 30 units. Find the income elasticity of demand.

$$\Delta Q = 5 \cdot \Delta Y = 100 \cdot Y = 100, Q = 25$$

$$E_y = (5 / 100) \times (100 / 25) = 0.2$$

$0 < 0.2 < 1 \rightarrow$ a **necessity (normal good)**. Income doubled (+100%) but quantity rose only 20%.

5. Promotional elasticity [L7 P3]

Quantity demanded increases from 8000 to 10,000 units because advertisement expenditure rises from ₹6000 to ₹12000. Find the promotional elasticity of demand.

$$\Delta Q = 2000 \cdot \Delta A = 6000 \cdot A = 6000, Q = 8000$$

$$E_a = (2000 / 6000) \times (6000 / 8000) = 0.25$$

Positive $\rightarrow$ the advertising **does** expand demand, so the promotional spend is beneficial (the deck's conclusion). It is < 1 , though: doubling the ad spend raised quantity by only 25%.

Traps

- **Original** values go in X/Q , not the new ones.
- Keep the **sign** for income and cross elasticity, because the sign *is* the answer. For price, report -2.5 and interpret it using $|E_p|$.
- Cross elasticity: the numerator is **X's** quantity and the denominator is **Y's** price.

Self-test

1. Price rises from ₹10 to ₹12 and quantity falls from 200 to 180. Find E_p and classify it.

2. Income rises from ₹20,000 to ₹25,000 and quantity falls from 10 to 9. Find E_y . What kind of good is it?
 3. The price of Y rises from ₹40 to ₹50 and the demand for X rises from 100 to 120. Find E_{xy} . How are X and Y related?
 4. Draw (in words) the demand curve for $E = 0$ and for $E = \infty$.
-

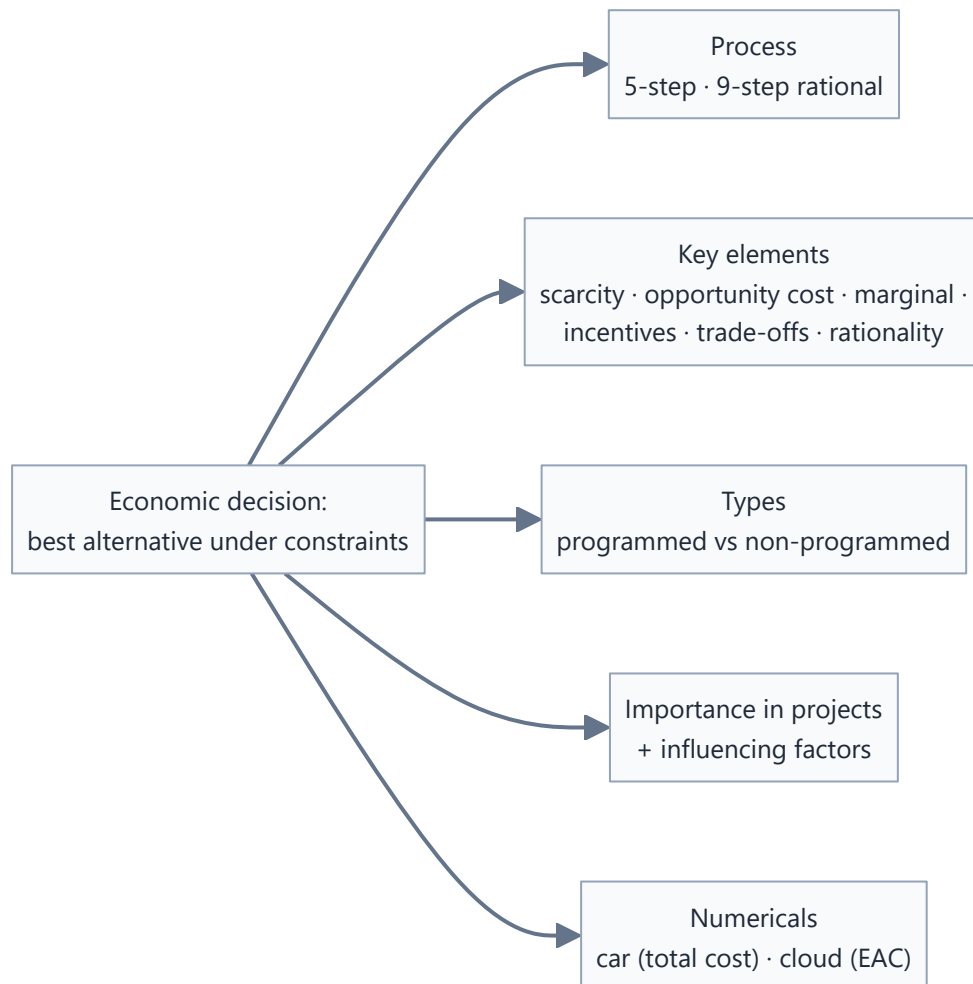
Answers

1. $(-20/2) \times (10/200) = -0.5 \rightarrow |E| < 1$, **relatively inelastic**.
2. $(-1/5000) \times (20000/10) = -0.4 \rightarrow$ negative, so an **inferior good**.
3. $(20/10) \times (40/100) = +0.8 \rightarrow$ positive, so **substitutes**.
4. $E = 0$: a **vertical** line (the same quantity at every price). $E = \infty$: a **horizontal** line (any price rise kills demand).

04 — Economic decision making (L8–L9)

Mostly theory, and long-answer friendly: "Explain the rational decision-making process" or "importance of economic decision making in engineering projects" are 5-mark questions answered by listing the points. There are two small numericals, and **one of them has a slide error**.

Map



1. Definition and process

Economic decision making: choosing, from the available alternatives, the one that **maximizes utility or benefit or minimizes cost** within the constraints. It rests on rational weighing of costs and benefits. Individuals, firms and governments all do it.

Decisions are made about **problems** (undesirable situations, e.g. cutting costs 10%) and **opportunities** (desirable ones, e.g. unexpected extra profit). Whether a decision was right may not be known for a long time.

Short process (5 steps): identify the problem → list the alternatives → evaluate costs and benefits → select the best → implement and review.

Rational decision-making process (9 steps). Memorize this order:

1. Recognize the problem
2. Define the goal or objective
3. Assemble relevant data (is more data worth its cost?)
4. Identify feasible alternatives
5. Select the criterion for choosing (economic, political, environmental, social, or a composite)
6. Mathematically model the interrelationships
7. Predict the outcome of each alternative
8. Choose the best alternative
9. Audit the results

Q ▶ (L8-9) Map buying a car (Audi vs BMW) onto the steps.

Need a car → a high-level security system → gather technical and financial data → alternatives: Audi, BMW
→ criterion: minimum total cash outlay → choose Audi (the deck's outcome).

2. Key elements

Element	Meaning	Deck example
Scarcity	limited resources, unlimited wants	a Govt can't build highways, hospitals and schools all at once
Opportunity cost	the value of the next-best alternative forgone	₹50,000 on a laptop = the skills/income from the course you skipped
Marginal analysis	compare the additional benefit and cost of one more unit	MC = 120, MB = 150 → MB > MC, so produce it
Incentives	rewards/penalties that change behaviour	higher wages → more effort. Tax rebates → more saving
Trade-offs	give up one thing to get another	quality vs cost. Defence vs welfare
Rationality	agents act logically to maximize utility/profit (not the same as selfishness)	buy the best satisfaction within budget, not necessarily the cheapest
Decision models	systematic frameworks	cost-benefit analysis, utility maximization, PPC

3. Types of decisions

	Programmed	Non-programmed
Nature	routine, repetitive, follows predefined rules	unstructured, rare, novel
Traits	structured, predictable, frequent, based on past experience	non-routine, complex, uncertain, needs judgment
Deck analogy	a train on a fixed track	a train at a junction where a human picks the direction
Examples	a McDonald's Big Mac procedure, Starbucks supply reordering	mergers, acquisitions, new facilities/products, labour contracts, legal issues
Who makes it	lower levels, by procedure	top management , using intuition and experience

4. Importance in engineering projects (list for a long answer)

1. **Cost control:** compare designs/materials, stay within budget without losing quality
2. **Risk management:** cost overruns, delays, price swings, with the best risk–reward option chosen
3. **Return on investment:** large capital, so pick the higher-ROI alternative
4. **Life cycle cost:** initial + operating + maintenance + replacement + disposal, not just first cost
5. **Sustainability:** include environmental/social costs
6. **Stakeholder alignment:** numbers give a common basis
7. **Project feasibility:** reject unviable projects early
8. **Resource optimization:** materials, labour, time, energy
9. **Competitiveness:** viable, high-return projects attract investors and clients

Factors influencing engineering economic decisions: cost–benefit analysis · budget constraints · technological feasibility · resource availability · life cycle costs · market demand and competition · regulatory and environmental rules · risk assessment · time constraints · government policies and incentives · stakeholder expectations · scalability and flexibility.

Case (L8-9): electronics firm, domestic vs international plant. Factors to weigh: labour cost vs shipping/logistics · market potential · regulatory environment · supply chain · political/economic stability · brand perception · financial viability (revenue, expenses, ROI in each scenario).

5. Petrol car vs electric car [L8-9 case]

	Petrol	Electric
Purchase	8,00,000	12,00,000
Running cost / yr	90,000	35,000
Life	8 yr	8 yr
Resale after 8 yr	2,00,000	3,00,000

Which car should the student buy based on total cost over 8 years?

Q ▶ Guess first: does the ₹4 lakh higher price get paid back?

	Petrol	Electric
Purchase	8,00,000	12,00,000
+ running × 8	7,20,000	2,80,000
– resale	2,00,000	3,00,000
Net 8-year cost	13,20,000	11,80,000

Electric, cheaper by ₹1,40,000. It saves ₹55,000/yr in running cost (₹4,40,000 over 8 years) plus ₹1,00,000 more at resale, and that outweighs the ₹4,00,000 higher price. (*The deck ignores the time value of money here. File 08 shows how you would discount it.*)

6. Cloud hosting: on-demand vs reserved [L8-9 example] slide error

A: on-demand, setup 0, ₹15,000/month. B: reserved, ₹2,50,000 upfront + ₹7,000/month. 3 years, interest 10% per year. Which is economically better?

Q ▶ Before reading on: B's upfront ₹2.5 lakh has to be spread over 3 years **with interest**. How?

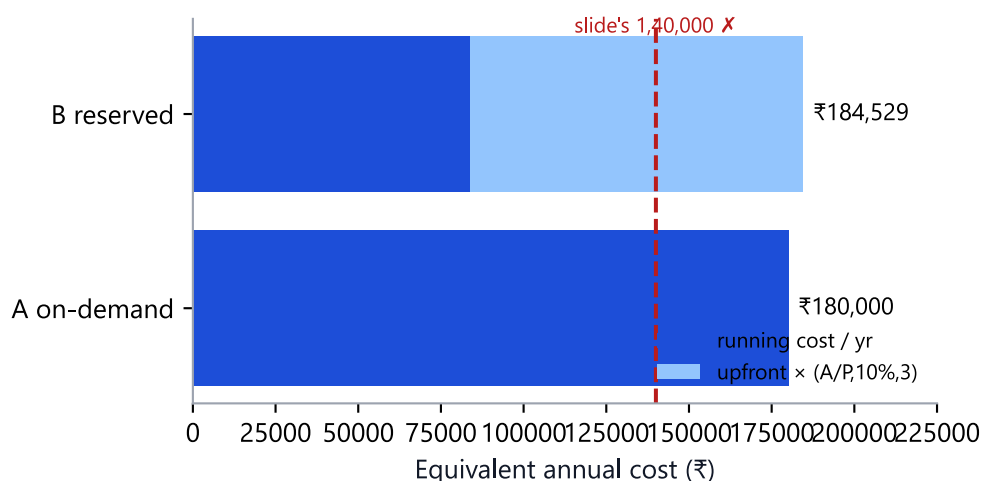
That is the **equivalent annual cost (EAC)**: convert every cost into an equal annual amount, then compare. An upfront P becomes an annual amount using the **capital recovery factor**:

$$(A/P, i, n) = i / (1 - (1+i)^{-n}) \quad (A/P, 10\%, 3) = 0.10 / (1 - 1.1^{-3}) = 0.40211$$

	A (on-demand)	B (reserved)
Running cost / yr	$15,000 \times 12 = 1,80,000$	$7,000 \times 12 = 84,000$
Upfront, annualized	0	$2,50,000 \times 0.40211 = 1,00,529$
EAC	₹1,80,000	₹1,84,529

At 10%, A (on-demand) is cheaper, by $\approx$ ₹4,529/yr.

 **The slide states $EAC(B) = ₹1,40,000$ and chooses B. That doesn't follow from its own data.** Even ignoring interest completely, $B = 84,000 + 2,50,000/3 = ₹1,67,333$, not 1,40,000. With no interest B would win, but at 10% interest A wins. (This treats the monthly costs as annual sums, which is the slide's simplification.) **In the exam:** show the working. If this exact question appears, write the computed answer and one line on the assumption.



Cloud hosting at 10%: B's upfront cost, annualized, makes B dearer than A.

Self-test

1. Define opportunity cost and give an engineering example.
2. List the 9 steps of the rational decision-making process.
3. Programmed vs non-programmed: give 3 differences and one example of each.
4. $MC = ₹200$, $MB = ₹180$. Produce the extra unit? Why?
5. What is the capital recovery factor $(A/P, i, n)$ used for?

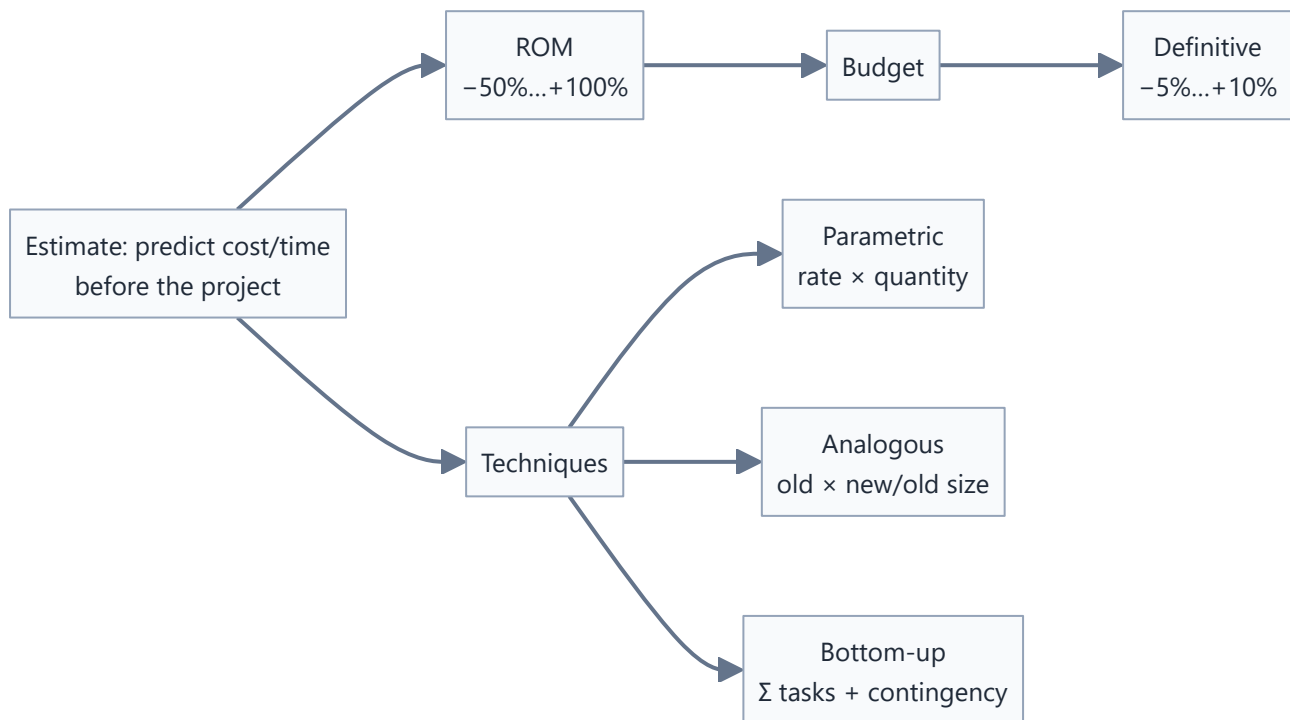
Answers

1. The value of the next-best alternative forgone. E.g. using a machine for product X means giving up the profit it would have made on product Y.
2. Recognize problem → define goal → assemble data → identify feasible alternatives → select criterion → model → predict outcomes → choose best → audit.
3. Routine vs novel. Rule-based vs judgment-based. Lower-level vs top management (also frequent vs rare). Examples: reordering supplies / a merger.
4. **No.** $MB < MC$, so the extra unit loses ₹20.
5. To convert a present (upfront) amount into an equivalent **equal annual** amount over n years at rate i . It is used for EAC comparisons.

05 — Types of estimates (L10–L12)

Theory plus one-line numericals. Expect "explain X estimate with an example" or "differentiate ROM and definitive". Each type has a one-line formula, so learn the formula with the definition.

Map



1. ROM (Rough Order of Magnitude) estimate [L10-12]

Q ▶ "How long from Delhi to Mumbai?" You answer "1–2 hours by flight, 20–25 by train" without checking a schedule. What kind of estimate is that?

A **ROM** (ballpark) estimate: a very high-level, deliberately imprecise estimate of cost, time or resources, made in the **earliest planning stage** from experience and limited information. **Accuracy** $\approx$ **–50% to +100%** (the slide's range).

Key points (write 4–5): **approximate** (often given as a range) · **limited detail** (it ignores the full scope and risks) · **quick** feasibility check · useful for **communicating** a first idea to stakeholders · a **basis for further planning** · **subject to change**, so it is replaced by better estimates later and is never the sole basis for critical decisions.

The ROM numericals (the method is to take the average of the range, then multiply):

Deck problem	Working	ROM
Office building, ₹150–200/sq ft, 10,000 sq ft	$(150+200)/2 = 175 \times 10,000$	₹17,50,000
Software project, similar ones took 6–9 months	$(6+9)/2$	7.5 months

Deck problem	Working	ROM
Gadget, ₹50–70/unit, 5,000 units	60 × 5,000	₹3,00,000
College web app: dev 25,000 + hosting 2,000 + tools 2,000	29,000	~ ₹30,000 , range 15,000–60,000

(The web-app range is –50% / +100% of 30,000.)

2. Budget estimate

More refined than ROM but still based on limited information. It is a **preliminary** assessment of the money needed, made **before detailed planning**, and it is used to judge feasibility and **set initial funding limits**.

Deck problem	Working	Budget
Building: materials 2,00,000 + labour 3,00,000 + equipment 1,00,000 + overhead 10% of direct	direct = 6,00,000. Overhead = 60,000	₹6,60,000
Software: R&D 1,50,000 + testing 80,000 + marketing 50,000 + misc 20,000	sum	₹3,00,000

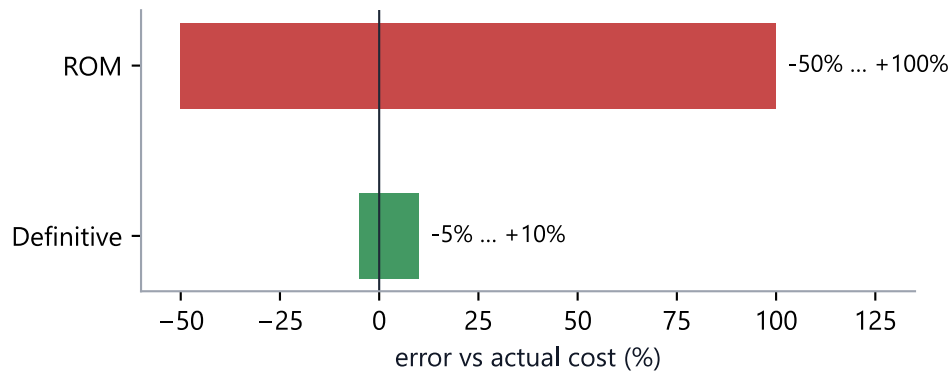
3. Definitive estimate

The **most accurate** estimate (–5% to +10%), produced **late**, once the scope, drawings and specifications are complete. It uses a comprehensive cost breakdown (materials, labour, equipment, overhead, contingency).

Characteristics (list for 5 marks): detailed scope · comprehensive plans and specs · accurate quantities · accurate cost breakdown · reliable data (current prices, history) · few, documented assumptions · low risk · high accuracy · used for **approval and funding** · final planning stage. Downsides: costly and slow to prepare, and unsuitable when the scope is still changing. Deck examples: an ERP implementation and an e-commerce app, both going scope → WBS → resources → cost → risk → schedule → QA → client review → final estimate.

ROM vs definitive: the likely "differentiate" question

	ROM	Definitive
When	earliest stage	final planning stage
Information	minimal	complete design and specs
Accuracy	–50% ... +100%	–5% ... +10%
Effort	minutes, from experience	large (detailed breakdown)
Used for	go/no-go, first idea	funding approval, contracts



Accuracy ranges from the slides: ROM vs definitive.

4. Parametric estimate

Q ▶ Houses around you cost ₹200/sq ft to build. Yours is 2,000 sq ft. Estimate it.

Cost = rate × quantity = 200 × 2,000 = **₹4,00,000**. That is **parametric** estimating: a **statistical relationship** between a parameter (sq ft, kg, components) and cost/time, taken from historical data, then applied to the new project's parameter value. It is quick, but it assumes the relationship still holds, so it is weak for unusual projects.

Deck problem	Parameter	Working	Estimate
Steel shafts: ₹180/kg, 250 shafts × 2.5 kg	weight	625 kg × 180	₹1,12,500
PCBs: ₹35 per component, 1,200 PCBs × 80 components	components	96,000 × 35	₹33,60,000

5. Analogous estimate [L10-12]

Previous CMS: 10,000 lines of code, 1,000 person-hours. New CMS: 15,000 lines. Estimate the effort.

Q ▶ Set up the ratio.

Analogous (top-down) estimating scales a **similar past project**:

$$\begin{aligned} \text{Estimated effort} &= \text{previous effort} \times (\text{new size} / \text{previous size}) \\ &= 1,000 \times 15,000 / 10,000 = 1,500 \text{ person-hours} \end{aligned}$$

Process: identify a reference project → gather its data → analyse similarities and differences → apply **adjustment factors** (e.g. + % for stricter regulation) → calculate. It is rough, and its accuracy depends on how similar the projects really are.

Parametric vs analogous: parametric uses a **statistical rate from many past data points** (₹/kg). Analogous scales **one comparable project** with judgment.

6. Bottom-up estimate [L10-12]

CMS tasks: requirements 40 h, database 20, UI 30, backend 300, frontend 250, testing 80, deployment 20.
Add a 20% contingency.

Q ▶ Total, then contingency.

$$\Sigma = 40 + 20 + 30 + 300 + 250 + 80 + 20 = \mathbf{740 \text{ hours}} \rightarrow \times 1.20 = \mathbf{888 \text{ hours}}$$

Bottom-up: break the project into the smallest tasks (a WBS), estimate each, and **aggregate**. It is the **most accurate** technique and the **most time-consuming**. Workflow: task identification → task estimation → aggregation → **contingency** → validation with stakeholders → documentation.

Self-test

1. Give the accuracy ranges of ROM and definitive estimates (as on the slides).
 2. A bridge costs ₹8 crore per km historically. The new one is 2.5 km. Name the technique and give the estimate.
 3. A past 20-storey tower took 30 months. Estimate a 30-storey one by analogy.
 4. Tasks of 12, 30 and 18 days plus a 15% contingency. Name the technique and give the estimate.
 5. Why should a ROM estimate never be the sole basis for a critical decision?
-

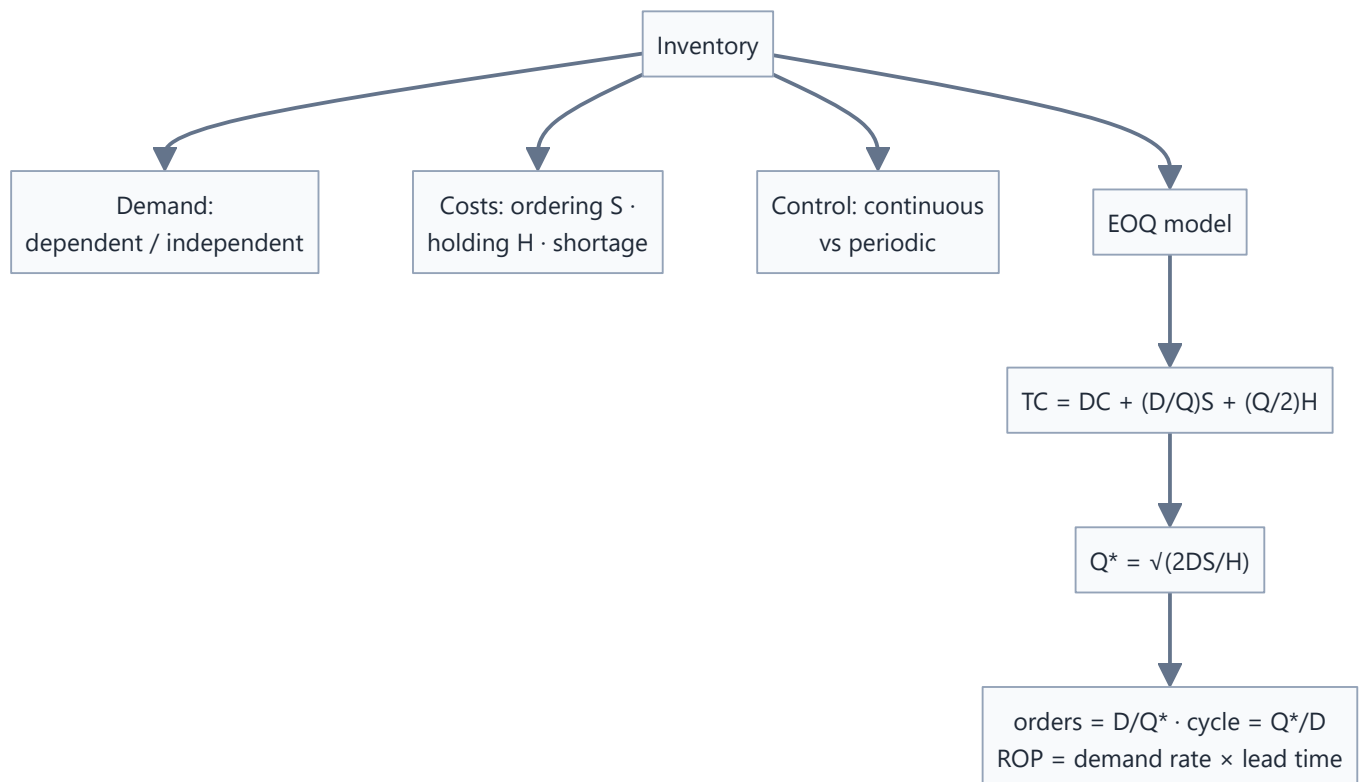
Answers

1. ROM –50% to +100%. Definitive –5% to +10%.
2. Parametric: $8 \times 2.5 = \text{₹20 crore}$.
3. Analogous: $30 \times 30/20 = \text{45 months}$ (then adjust for differences).
4. Bottom-up: $60 \times 1.15 = \text{69 days}$.
5. It is based on minimal information and assumptions and ignores the full scope and risks, so the actual figure can be twice the estimate.

06 — Inventory and EOQ (L13 + the EOQ practice sheet)

High-yield numerical. The professor handed out a practice sheet for this lecture, which is the nearest thing to an assignment you have. **All 4 sheet questions and both deck examples are answered here, and every number was recomputed.**

Map



1. The words

- **Inventory:** a stock of items kept to meet future demand (the shop's 50 phones). Managing it means answering **how many to order** and **when to order**.
- **Dependent demand:** items used to make a final product (tyres for cars). **Independent demand:** items bought by external customers (cars, appliances, houses).
- **Costs:** **ordering cost S** (the cost of placing/replenishing one order) · **holding/carrying cost H** (the cost of keeping one unit in stock for a year) · **shortage cost** (sales lost when demand can't be met).
- **Continuous (fixed-order-quantity) system:** order a **fixed amount** when stock falls to a set level ("at 50 sugar packets, reorder"). **Periodic (fixed-time-period) system:** check at **fixed intervals** and order a **variable** amount ("check monthly, top up").

2. The EOQ model

Q ▶ Order in huge batches and you rarely pay the ordering cost, but you store a lot. Order in tiny batches and storage is cheap, but you pay the ordering cost constantly. Where is the minimum?

EOQ (Economic Order Quantity) is the order size Q that **minimizes total inventory cost**.

Assumptions (a common 2-marker): demand known, constant and deterministic · ordering cost per order fixed · holding cost per unit per year constant · instant replenishment (zero lead time) · no shortages · constant unit price.

Building the total cost (D = annual demand, C = unit price, S = cost per order, H = holding cost/unit/yr):

Purchasing cost = $D \times C$	(fixed, since you buy D whatever Q is)
Ordering cost = $(D / Q) \times S$	number of orders $\times$ cost per order
Holding cost = $(Q / 2) \times H$	stock falls uniformly $Q \rightarrow 0$, so the average is $Q/2$
$TC = DC + (D/Q)S + (Q/2)H$	

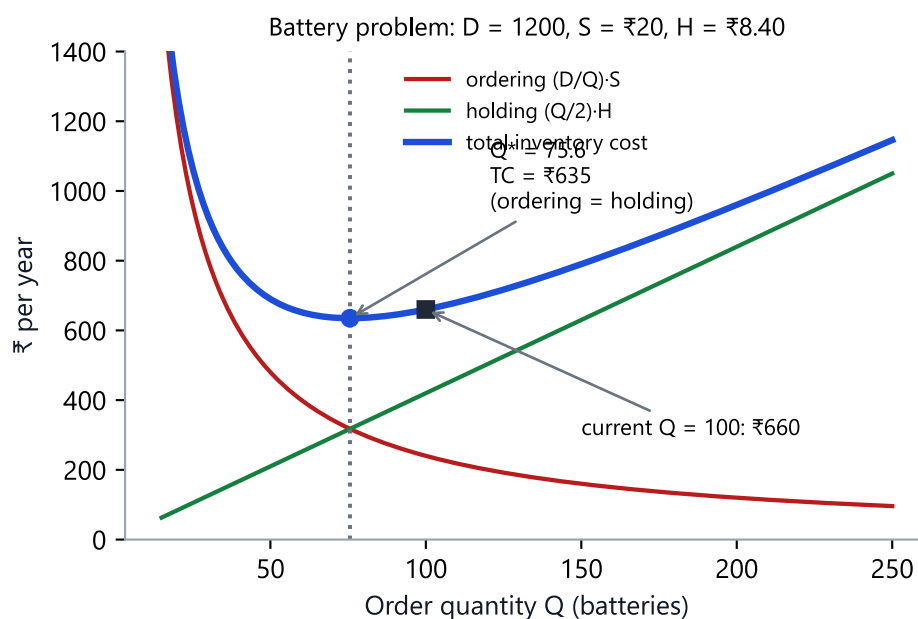
Derivation (write it for a 5-marker):

$$dTC/dQ = -DS/Q^2 + H/2 = 0 \Rightarrow Q^2 = 2DS/H \Rightarrow Q^* = \sqrt{(2DS / H)}$$

At Q^* , ordering cost = holding cost. That is the trade-off picture: the ordering cost falls as Q rises, the holding cost climbs, and the total cost is U-shaped with its bottom where the two cross.

After Q^* :

number of orders / yr = D / Q^*	
cycle time = Q^* / D years	($\times 12$ for months, $\times 365$ for days)
reorder point (ROP) = demand rate $\times$ lead time	(lead time = order placed $\rightarrow$ order received)
H given as a %?	$H = \% \times \text{unit price}$



The EOQ trade-off, drawn with the battery problem's numbers (\$5).

3. Car dealer [L13 example]

Demand 5,000 cars/yr. It costs 15,000 to have cars shipped (per order). Holding cost 500 per car per year. How many times should the dealer order, and what should the order size be?

Q ▶ Identify D, S and H first.

$$D = 5,000 \cdot S = 15,000 \cdot H = 500$$

$$Q^* = \sqrt{(2 \times 15,000 \times 5,000 / 500)} = \sqrt{3,00,000} = \mathbf{548 \text{ cars per order}}$$

$$\text{Orders/yr} = 5,000 / 548 = \mathbf{9.13 \approx 9 \text{ times a year}}, \text{ roughly every 40 days.}$$

4. Raw material, with carrying cost as a % [L13 example]

16,000 units/yr at ₹2 per unit. Ordering cost ₹45. Carrying cost 10% per year per unit of the average inventory. Find the EOQ and the cycle time.

Q ▶ H isn't given in rupees. Get it first.

$$H = 10\% \times ₹2 = \mathbf{₹0.20} \text{ per unit per year}$$

$$Q^* = \sqrt{(2 \times 16,000 \times 45 / 0.20)} = \sqrt{72,00,000} = \mathbf{2,683 \text{ units}}$$

$$\text{Orders/yr} = 16,000 / 2,683 = \mathbf{5.96 \approx 6}$$

$$\text{Cycle time} = 2,683 / 16,000 = 0.168 \text{ yr} = \mathbf{\approx 2 \text{ months (61 days)}}$$

5. Batteries [EOQ sheet Q1]

Annual demand $\approx$ 1,200 batteries. Cost ₹28 each. Holding cost 30% of the battery's cost. ₹20 per order. The supplier currently orders 100 per month. (a) inventory cost of the current order quantity (b) EOQ (c) orders per year at the EOQ.

$$H = 0.30 \times 28 = \mathbf{₹8.40}, S = 20, D = 1,200$$

(a) $Q = 100$: ordering = $(1200/100) \times 20 = \mathbf{₹240}$. Holding = $(100/2) \times 8.40 = \mathbf{₹420}$. **Total ₹660/yr** (the sheet writes $(1200 \times 100) / 20$ here. It means $(1200/100) \times 20$, and the result, 240, is right.)

$$\mathbf{(b)} \quad Q^* = \sqrt{(2 \times 1200 \times 20 / 8.40)} = \sqrt{5714.3} = \mathbf{75.6 \approx 76 \text{ batteries}}$$

$$\mathbf{(c)} \quad 1200 / 75.6 = \mathbf{15.9 \approx 16 \text{ orders/yr}}$$

At the EOQ the cost is $317.5 + 317.5 = \mathbf{₹635/yr}$, which saves ₹25/yr over the current policy. Ordering = holding, as predicted.

6. Indian Telecom, with a reorder point [EOQ sheet Q2]

Indian Telecom contracts to buy 12,500 instruments during the year at ₹2.50 each. "The deliveries of the instruments will be made each time twice a month after the order is placed." Carrying cost ₹48 per instrument per annum. Paperwork, follow-up, transport and receipt work out to ₹2,000. How frequently should orders be placed? What is the re-order point?

Q ▶ Which of the given numbers is S, and which is H?

$$S = ₹2,000 \text{ (per order)} \cdot H = ₹48 \cdot D = 12,500$$

$$EOQ = \sqrt{(2 \times 12,500 \times 2,000 / 48)} = \sqrt{10,41,667} = \mathbf{1,021 \text{ instruments}}$$

$$\text{Orders/yr} = 12,500 / 1,021 = \mathbf{12.2 \approx 12} \text{ (about once a month)}$$

Lead time: the sheet reads the garbled delivery sentence as $\frac{1}{2}$ month = $\frac{1}{24}$ yr (state that assumption) → **ROP = $12,500 / 24 = 521$ units**. Reorder when the stock falls to 521. (The ₹2.50 unit price doesn't enter the EOQ, because H is already given in rupees.)

7. Lead time in months [EOQ sheet Q3]

Consumption 20 units/yr. Procurement cost ₹40 per order. Unit cost ₹100. Carrying cost 16% of the average stock. (i) EOQ
(ii) with a lead time of 3 months, the reorder point.

$$H = 0.16 \times 100 = \mathbf{₹16} \text{ (the sheet prints "0.16\%". It means 16\%, because 0.16\% would make H ₹0.16 and the EOQ 100.)}$$
$$EOQ = \sqrt{(2 \times 20 \times 40 / 16)} = \sqrt{100} = \mathbf{10 \text{ items}}$$
$$\text{Consumption rate} = 20/12 \text{ per month} \rightarrow \text{ROP} = (20/12) \times 3 = \mathbf{5 \text{ units}}$$

8. Bicycle tyres [EOQ sheet Q4]

450 bicycles a month. Tyres at ₹20 each. Carrying cost 15% of cost. Ordering cost ₹50 per order. (i) EOQ (ii) orders per year
(iii) total cost.

Q ▶ What is D? (Look at "bicycles" and "tyres" again.)

$$D = 2 \text{ tyres} \times 450 \times 12 = \mathbf{10,800 \text{ tyres/yr (the trap: 2 tyres per bicycle, and monthly} \rightarrow \text{annual)}}$$
$$H = 0.15 \times 20 = \mathbf{₹3}, S = 50$$
$$EOQ = \sqrt{(2 \times 10,800 \times 50 / 3)} = \sqrt{3,60,000} = \mathbf{600 \text{ tyres}}$$
$$\text{Orders} = 10,800 / 600 = \mathbf{18 \text{ per year (iii)}}$$
$$\text{Ordering} = 18 \times 50 = ₹900. \text{ Holding} = (600/2) \times 3 = ₹900 \rightarrow \text{inventory cost } \mathbf{₹1,800}.$$

Including purchase ($10,800 \times 20 = ₹2,16,000$): **TC = ₹2,17,800/yr.** (The sheet stops before (iii). Give both figures and label them.)

Traps

- **H must be in ₹ per unit per year.** If it is given as a %, multiply by the unit price.
- **D must be annual.** Convert monthly figures, and check units per product (tyres per bike).
- The EOQ never depends on the purchase price C directly, only through H when H is a %.

- ROP needs the lead time in the **same time unit** as the demand rate.
 - Round the EOQ to whole units, and say so.
-

Self-test

1. $D = 2,400/\text{yr}$, $S = ₹50$, $H = ₹6$. Find the EOQ, orders/yr, cycle time, and the total inventory cost.
 2. Same item, lead time $\frac{1}{2}$ month. Find the ROP.
 3. Show that ordering cost = holding cost at Q^* .
 4. State 4 assumptions of the EOQ model.
-

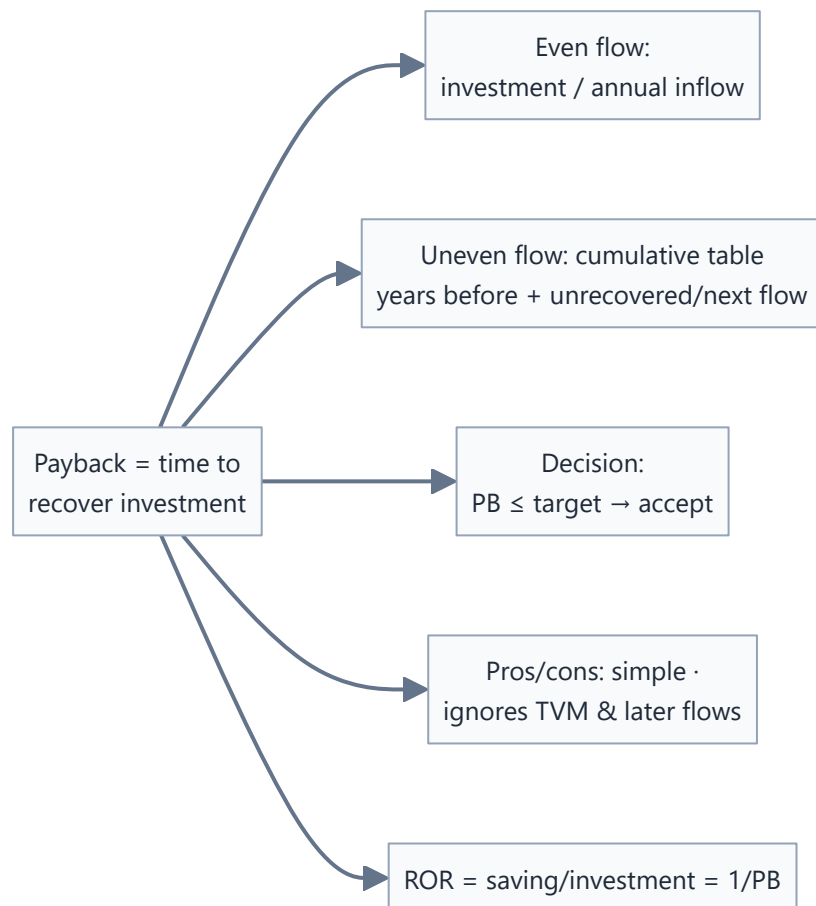
Answers

1. $Q^* = \sqrt{(2 \times 2400 \times 50 / 6)} = \sqrt{40,000} = \mathbf{200}$. Orders = **12/yr**. Cycle = $200/2400$ yr = **1 month**. Cost = $12 \times 50 + 100 \times 6 = 600 + 600 = \mathbf{\text{₹}1,200}$.
2. $ROP = (2400/12) \times \frac{1}{2} = \mathbf{100 \text{ units}}$.
3. Put $Q^* = \sqrt{(2DS/H)}$ into $(D/Q)S$: $D \cdot S / \sqrt{(2DS/H)} = \sqrt{(DSH/2)}$. And $(Q/2)H = \sqrt{(2DS/H)} \cdot H/2 = \sqrt{(DSH/2)}$. They are equal.
4. Demand known and constant · fixed cost per order · constant holding cost · instant replenishment / no shortages · constant price.

07 — Simple payback and rate of return (L14)

Easy marks. There are two formulas, one table method, and one decision line. The deck, the Payback deck and the handwritten notes all use the same 5 problems, so they are very likely to come up.

Map



1. Payback period: even cash flow

Q ▶ [L14 · Payback deck] Company X's project costs \$10,500 and returns \$2,500 a year for 7 years. When does it pay for itself?

Payback period: the length of time needed to **recover the initial investment** from the project's net cash inflows. When every year's inflow is the same:

$$\text{Payback} = \text{initial investment} / \text{annual net cash inflow} = 10,500 / 2,500 = 4.2 \text{ years}$$

Notes version [handwritten notes p1]: a lighting retrofit costs \$1,000 and saves \$250/yr → SP = $1000/250 = 4$ years.

2. Net inflow first, then payback [L14 · beverage equipment]

Equipment \$37,500, life 10 years, maximum desired payback 4 years. Annual sales \$75,000. Annual outflows: ingredients \$45,000, salaries \$13,500, maintenance \$1,500. Should XYZ Beverage buy it (payback method)?

Q ▶ What goes in the denominator: the sales, or something else?

The **net** annual inflow:

Net = 75,000 – (45,000 + 13,500 + 1,500) = **\$15,000**

Payback = 37,500 / 15,000 = **2.5 years**

Decision: 2.5 < 4 (the maximum desired) → **purchase the equipment.**

3. Uneven cash flow: the cumulative table [L14]

Investment \$200,000. Inflows: Y1 70,000 · Y2 60,000 · Y3 55,000 · Y4 40,000 · Y5 30,000 · Y6 25,000. Find the payback. Invest if management wants recovery within 3 years?

Q ▶ Build the cumulative column and find the year the total crosses 200,000.

Year	Inflow	Cumulative
1	70,000	70,000
2	60,000	1,30,000
3	55,000	1,85,000 ← still short
4	40,000	2,25,000 ← crosses 2,00,000
5	30,000	2,55,000
6	25,000	2,80,000

Payback = years before full recovery + (unrecovered cost at start of the year / cash flow during that year)

$$= 3 + (200,000 - 185,000) / 40,000 = 3 + 0.375 = 3.375 \text{ years}$$

Decision: 3.375 > 3 → **don't invest** (the 3-year target is missed).

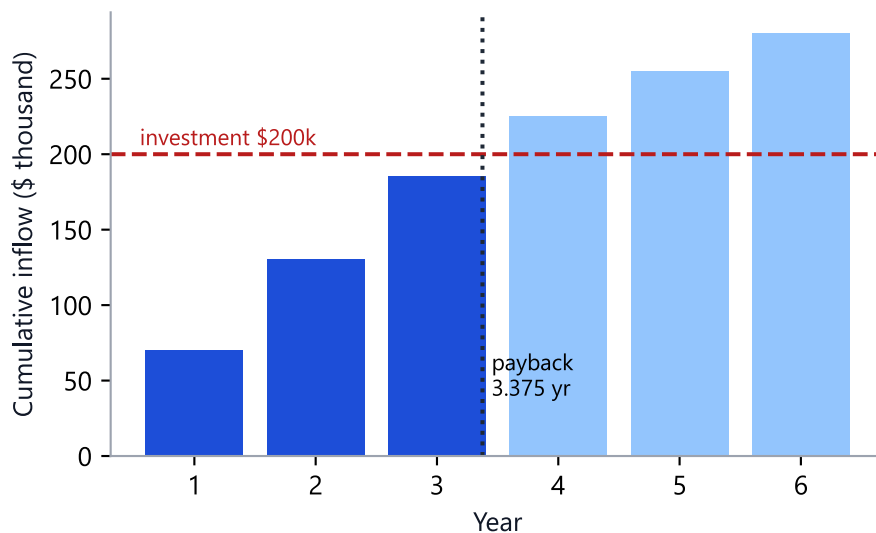
The unrecovered-balance version [L14 · notes p1]:

Investment ₹50,000. Cash flows ₹15,000, ₹20,000, ₹14,000, ₹16,000 over the next 4 years.

Year	Cash flow	Balance still to recover
0	–50,000	–50,000
1	15,000	–35,000
2	20,000	–15,000

Year	Cash flow	Balance still to recover
3	14,000	-1,000
4	16,000	+15,000 ← recovered during year 4

Payback = 3 + 1,000 / 16,000 = **3.0625 years**



The cumulative inflow crosses the \$200k investment during year 4.

4. Pros and cons (2–3 marks)

Pros	Cons
very simple and cheap to compute	ignores the time value of money
links annual cash flow directly to the investment	ignores everything after payback , and so ignores overall profitability
favours quick cash return, which is useful for cash-poor firms	

(File 08 \$6 shows the second con in action: two projects with nearly the same payback have very different NPVs.)

5. Rate of return (ROR) [L14]

Q ▶ For the \$10,500 / \$2,500 project, what fraction of the investment comes back each year?

ROR = the annual return on the investment, which is the **reciprocal of simple payback**:

ROR = saving per year / initial investment = 2,500 / 10,500 = 0.2381 = 23.8 %
 check: 1 / 4.2 = 0.238 ✓

Lighting retrofit (notes): 250 / 1000 = **25%** (= 1/4).

Traps

- Use **net** inflow (inflow – outflows), never gross sales.
 - In the fraction, the denominator is the **cash flow of the year in which recovery happens**, not the last year's.
 - Write the **decision** against the stated target.
-

Self-test

1. Cost ₹48,000, saving ₹12,000/yr. Find the payback and the ROR.
 2. Cost ₹60,000. Flows ₹20,000, ₹25,000, ₹30,000. Find the payback.
 3. Why can payback pick the wrong project?
-

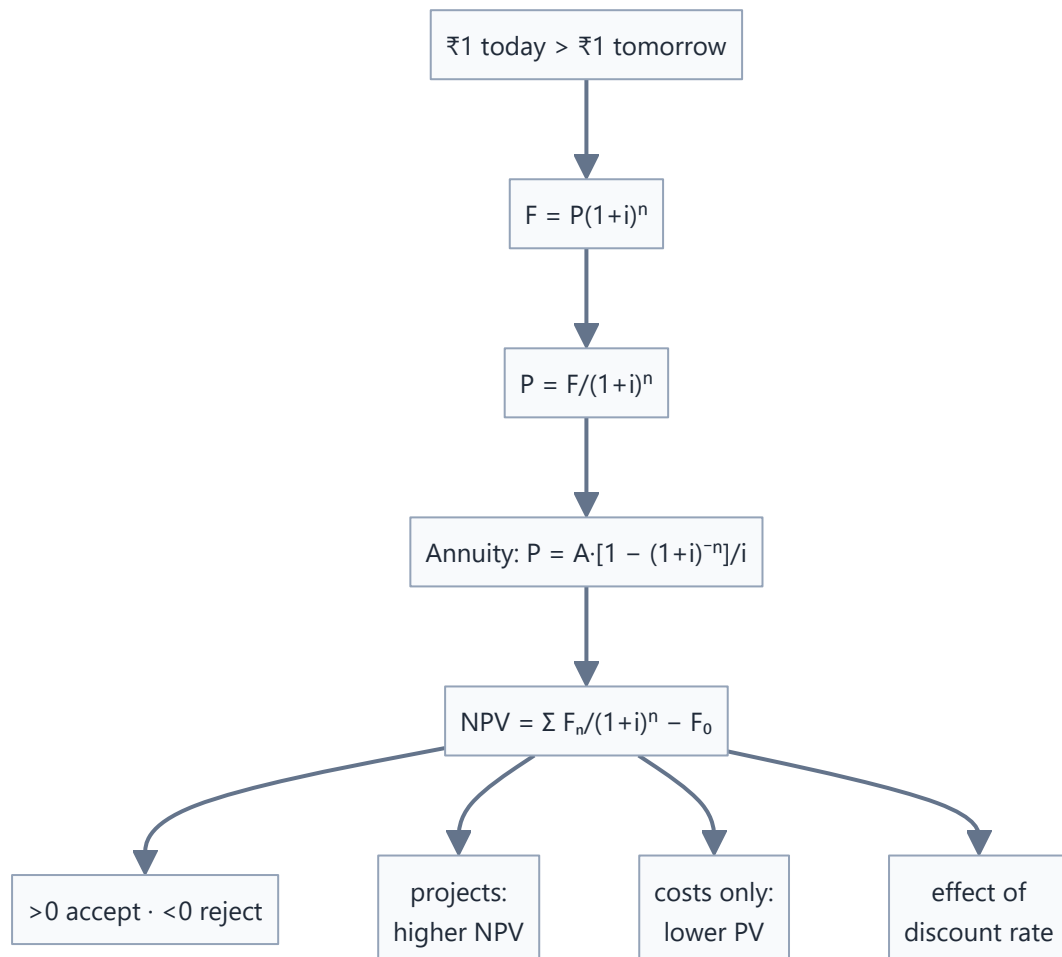
Answers

1. PB = **4 years**. ROR = $12,000/48,000 = 25\%$.
2. After year 2: 45,000 recovered, 15,000 left $\rightarrow 2 + 15,000/30,000 = 2.5$ **years**.
3. It ignores the time value of money and every cash flow after payback, so a project with big later returns (higher NPV) can lose to one that merely recovers faster.

08 — Time value of money and NPV (L15–L17)

The single most important file. NPV takes up 3 lectures and appears in the deck, in the older Payback deck, and in the handwritten notes. There are **8 professor problems here, all recomputed**. One of them has a **⚠ slide error**.

Map



1. Why money has a time value

Q ▶ [L15] Grandfather offers ₹1,00,000 at the end of year 5, or ₹75,000 today. Which do you take?

₹75,000 today (the deck's answer): the present is **certain**, ₹75,000 can be **invested to earn a return**, and inflation erodes ₹1,00,000 by year 5. Put a number on it: at 10%, ₹1,00,000 in 5 years is worth only $1,00,000/1.1^5 = \text{₹62,092 today}$, which is less than ₹75,000. (₹75,000 grows to ₹1,00,000 at only 5.9% a year, so the lakh wins only if you can't earn even 5.9%.)

2. Future value and present value

Compound interest, built step by step (the deck and notes use $P = 100$, $i = 5\%$):

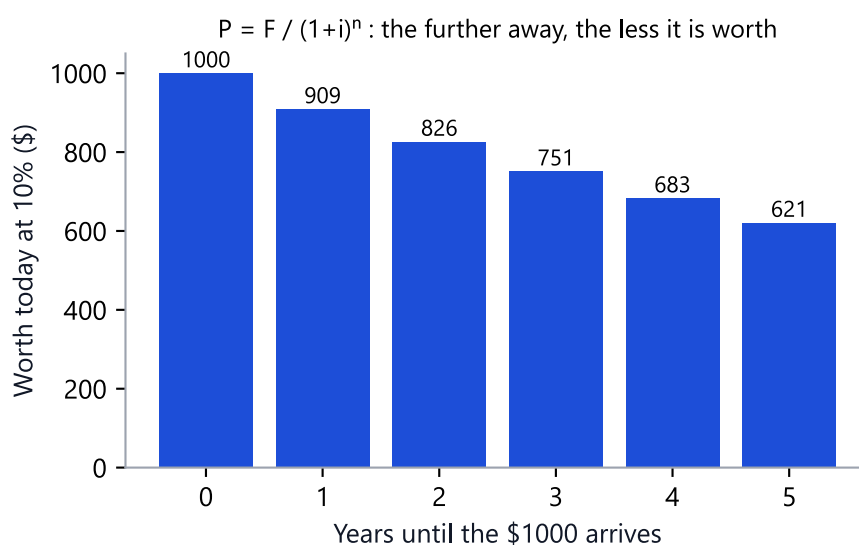
$$\begin{aligned}
 F_1 &= P + Pi &= P(1+i) & 100 \times 1.05 &= 105 \\
 F_2 &= F_1 + F_1 i &= P(1+i)^2 & 100 \times 1.05^2 &= 110.25 \\
 &\vdots &&& \\
 F_n &= P(1+i)^n &&& \text{(simple interest would be } F = P(1 + ni))
 \end{aligned}$$

Rearranged, it gives the **present value** of a future amount:

$$P = F / (1+i)^n = F(1+i)^{-n} \quad (1+i)^{-n} \text{ is the PRESENT WORTH FACTOR (PWF)}$$

Q ▶ [L15 · notes p3] Someone promises you \$1000 in 5 years. At 10%, what amount today is equivalent?

$$P = 1000 / 1.1^5 = 1000 / 1.61051 = \text{\$620.92} \approx \text{\$621}$$



\$1000 received n years from now, valued today at 10%.

3. Present value of a series (annuity)

If the same amount A arrives at the end of each of n years:

$$\begin{aligned}
 P_n &= A/(1+i) + A/(1+i)^2 + \dots + A/(1+i)^n \\
 &= A \cdot [1 - (1+i)^{-n}] / i \quad \text{(sum of a geometric progression with ratio } 1/(1+i))
 \end{aligned}$$

The bracket is the **annuity factor (P/A, i, n)**. Values used in this pack:

i, n	factor	i, n	factor
10%, 5	3.7908	10%, 10	6.1446
12%, 3	2.4018	12%, 8	4.9676
15%, 5	3.3522	20%, 5	2.9906
30%, 10	3.0915		

4. NPV: definition and decision rule

Net present value = PV of cash inflows – PV of cash outflows:

$$NPV = \sum_{n=1}^t F_n / (1+i)^n - F_0 \quad F_0 = \text{initial investment (the cash outlay at time 0)}$$

NPV	Meaning	Decision
> 0	inflows are worth more than the outlay, today	accept
< 0	the project destroys value	reject
= 0	it exactly earns the discount rate	indifferent (may accept or reject)

The discount rate **i** is the **expected return from the alternative investment** (the cost of capital). A **higher i** shrinks far-future cash flows more. (Deck example: Company X valuing Company A discounts A's 10-year projected cash flows at its WACC, the weighted average cost of capital, and subtracts the purchase price.)

5. Single project [L16-17 · notes p8]

Initial investment \$2000. Cash flow \$100 for 3 years plus \$2500 in the third year. Target rate 10%. Find the NPV.

Q ▶ Write the four discounted terms.

$$\begin{aligned} NPV &= 100/1.1 + 100/1.1^2 + 100/1.1^3 + 2500/1.1^3 - 2000 \\ &= 90.91 + 82.64 + 75.13 + 1878.29 - 2000 \\ &= \$126.97 > 0 \rightarrow \text{accept} \end{aligned}$$

6. Choosing between two projects [L16-17] ⚠ slide error

Initial investment \$10,000 each. Cost of capital 10%. Cash flows years 1–4: A = 5000, 4000, 3000, 1000 · B = 1000, 3000, 4000, 6750. Find the NPV of each. Which project should the firm choose?

Year	A	PV of A	B	PV of B
1	5000	4545.45	1000	909.09
2	4000	3305.79	3000	2479.34
3	3000	2253.94	4000	3005.26
4	1000	683.01	6750	4610.34
Σ PV		10,788.20		11,004.03
NPV		\$788.20		\$1,004.03

NPV(B) > NPV(A) → choose B.

⚠ The slide shows NPV A = \$780.18 (its PV sum is mis-added) and NPV B = \$1052.19 (it gives B's year-3

PV as 3053.43, which should be $4000/1.331 = 3005.26$). The conclusion is the same. Use the corrected figures.

7. Comparing costs: furnaces [L16-17 · notes p4-5]

A standard furnace costs \$100 and uses \$40/yr of fuel for 10 years. A high-efficiency furnace costs \$200 and uses \$20/yr for 10 years. At 10%, which is the better investment?

Q ▶ These are all costs, so which PV do you want: the higher or the lower?

The **lower** PV of total cost:

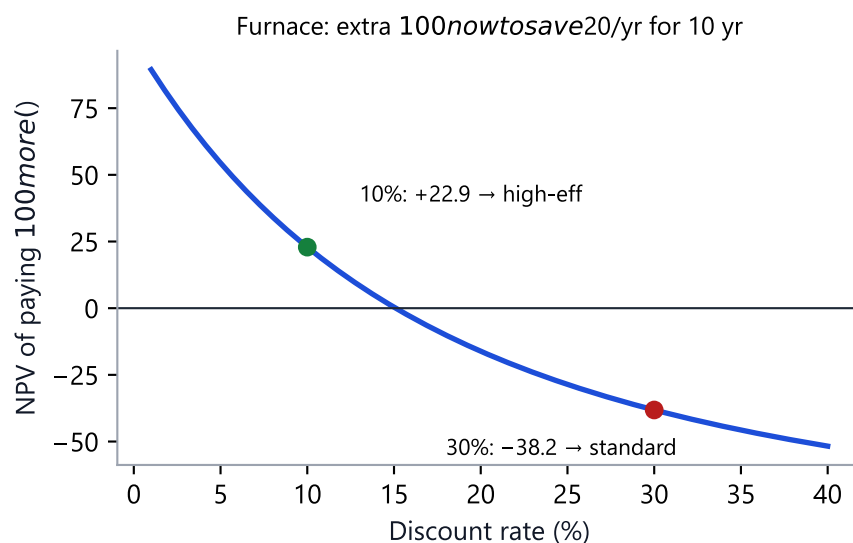
$$PV(\text{standard}) = 100 + 40 \times 6.1446 = \$345.78 \approx \$346$$

$$PV(\text{high-eff}) = 200 + 20 \times 6.1446 = \$322.89 \approx \$323$$

→ high-efficiency is cheaper by $\approx \$23$

Or directly (incremental NPV): pay \$100 extra, save \$20/yr → $-100 + 20 \times 6.1446 = +\$22.89 > 0$ → buy the high-efficiency one.

Effect of the discount rate. At 30%: $-100 + 20 \times 3.0915 = -\$38.17 < 0$ → the **standard** furnace wins. A high discount rate means the money could earn a lot elsewhere, so future savings count for little and the option with the **lower initial cost** is preferred.



The furnace decision flips as the discount rate rises (the crossover is $\approx 15\%$).

8. Payback and NPV disagree [L16-17 · notes p6]

Discount rate 10%. Year 0–5: A = -1200, 400, 400, 400, 200, 200 · B = -1200, 400, 400, 350, 800, 800. Evaluate by payback and by NPV, and select one.

Payback: A recovers 1200 exactly at the end of year 3 → **3 years**.

B: $400 + 400 + 350 = 1150$ after year 3, 50 left → $3 + 50/800 = 3.0625$ years.

NPV:

$$A = -1200 + 400/1.1 + 400/1.1^2 + 400/1.1^3 + 200/1.1^4 + 200/1.1^5 = \$55.53$$

$$B = -1200 + 400/1.1 + 400/1.1^2 + 350/1.1^3 + 800/1.1^4 + 800/1.1^5 = \$800.32$$

Select B. Payback prefers A by about 3 weeks (0.0625 yr), but ignores B's large years 4–5 flows. NPV counts them. This is the "ignores flows after payback" con made concrete.

9. Real estate [L16-17 practice problem]

Land costs ₹2,00,000. It generates ₹30,000/yr for the next 3 years. Cost of capital 12%. Find the NPV. Viable?

$$PV = 30,000 \times 2.4018 = \text{₹}72,055$$

$$NPV = -2,00,000 + 72,055 = \text{₹} -1,27,945 < 0 \rightarrow \text{not financially viable}$$

Notes version (8 years): $PV = 30,000 \times 4.9676 = \text{₹}1,49,029 \rightarrow NPV = \text{₹} -50,971$, still not viable. Check which duration your paper states.

10. Straight annuity [notes p9]

Initial investment ₹1,50,000, cash flow ₹50,000 each year, discount rate 10%, 5 years.

$$NPV = -1,50,000 + 50,000 \times 3.7908 = -1,50,000 + 1,89,539 = \text{₹}39,539 > 0 \rightarrow \text{accept}$$

Traps

- **Year 0 is not discounted.** The investment enters at full value.
- For **costs-only** comparisons (furnace), pick the **lowest** PV. For income projects, pick the **highest** NPV.
- Use the annuity factor only when the flows are **equal** every year. Otherwise discount each year separately.
- Show one line of substitution before jumping to the factor value.

Self-test

1. ₹5,000 invested at 8% compound for 3 years. Find F.
2. PV of ₹10,000 received in 4 years at 12%.
3. Cost ₹20,000, returns ₹6,000/yr for 5 years, $i = 10\%$. NPV and decision.
4. In one line: why does raising the discount rate favour the low-first-cost option?

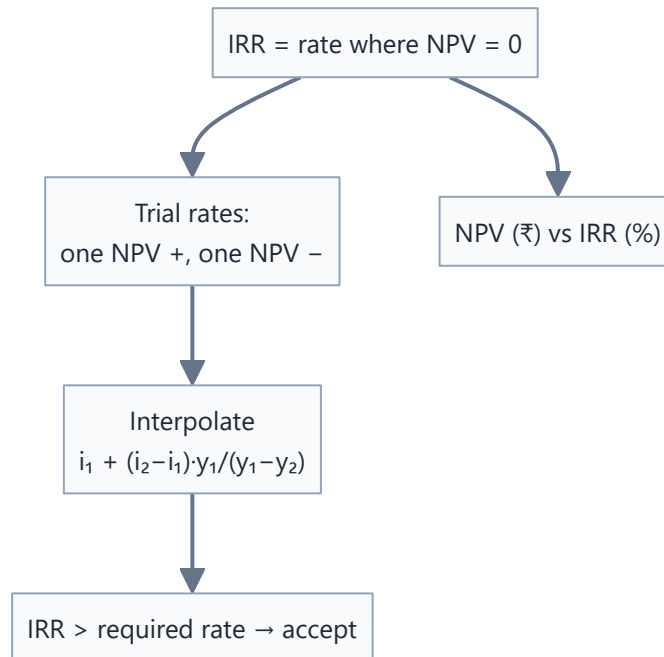
Answers

1. $F = 5000 \times 1.08^3 = \text{₹}6,298.56$
2. $P = 10,000 / 1.12^4 = \text{₹}6,355.18$
3. $NPV = -20,000 + 6,000 \times 3.7908 = \text{₹}2,744.72 > 0 \rightarrow \text{accept}$
4. A higher rate discounts distant savings more heavily, so the extra money spent up front is recovered in smaller present-value terms.

09 — Internal rate of return (L18–L19)

Numerical, and it builds directly on 08. IRR is the rate that makes NPV zero. You can't solve for it algebraically, so the exam method is **two trial rates + linear interpolation**. There are 3 professor problems here. One has a **Δ slide error** in its working.

Map



1. Definition

Q ▶ [L18] At 15% a machine's NPV is +\$282. At 20% it is -\$5,141. Somewhere in between, NPV is exactly zero. What is that rate called, and what does it tell you?

Internal rate of return (IRR): the **discount rate at which NPV = 0**, i.e. the rate at which the PV of the inflows equals the initial outlay. It is the project's **own annual rate of return** (its "yield"), and it is **not a rupee amount**.

$$0 = \sum_{n=1}^t \frac{F_n}{(1 + \text{IRR})^n} - P \quad \quad P = \text{initial investment}$$

Decision: IRR > the required rate (cost of capital) → **accept**. IRR is also the break-even discount rate: at that rate the project makes neither profit nor loss in present-value terms.

NPV vs IRR (a 2–3 marker):

NPV	IRR
PV(inflows) – PV(outflows)	the rate at which that difference is 0
a money amount	a percentage
needs the discount rate as an input	is the output. Compare it with the required rate

NPV	IRR
shows the excess of inflows over outflows	shows the break-even rate

(Notes p11: Excel's IRR assumes equally spaced periods. XIRR lets you give each cash flow its own date, which makes it more accurate for irregular timing.)

2. Interpolation

$$i^* = i_1 + (i_2 - i_1) \times (y^* - y_1) / (y_2 - y_1)$$

i_1 = rate with POSITIVE NPV y_1 i_2 = rate with NEGATIVE NPV y_2 $y^* = 0$
equivalently $i^* = i_1 + (i_2 - i_1) \times y_1 / (y_1 - y_2)$

Q ▶ [L18-19 demo] NPV is +200 at 10% and -100 at 15%. Find the IRR.

$$i^* = 10 + (15 - 10) \times (0 - 200) / (-100 - 200) = 10 + 5 \times 200 / 300 = \mathbf{13.33\%}$$

The two trial rates **must bracket** zero, one positive and one negative. Keep them close (within 5%), because interpolation assumes a straight line and the true NPV curve bends.

3. Machine [L18-19 · notes p11]

Initial cost - \$50,000. Additional \$15,000 each year for 5 years. Economic acceptability at (a) 10% (b) 15% (c) 20%. Also find the IRR.

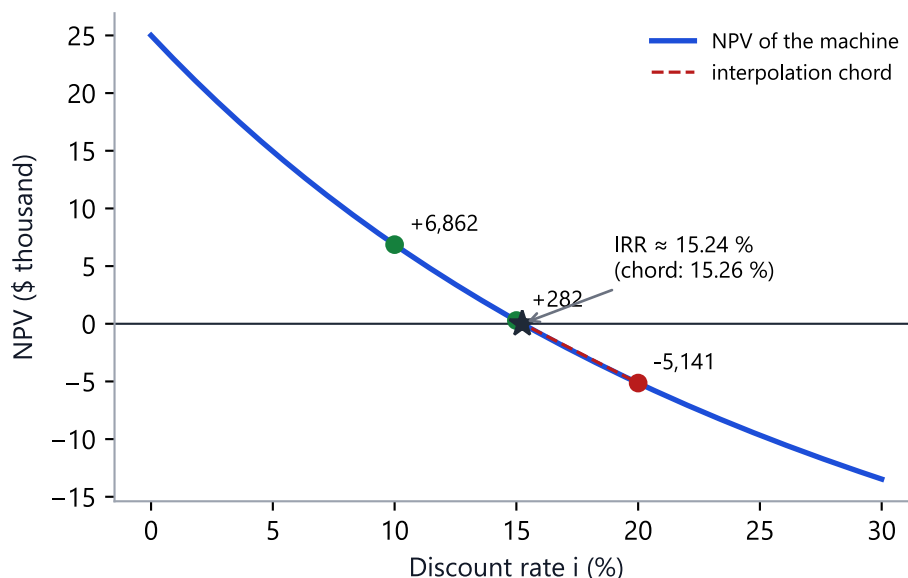
Q ▶ One annuity factor per rate. Which of the three NPVs will be negative?

(a) 10%: $-50,000 + 15,000 \times 3.7908 = +\$6,861.80$ acceptable
(b) 15%: $-50,000 + 15,000 \times 3.3522 = +\282.33 barely acceptable
(c) 20%: $-50,000 + 15,000 \times 2.9906 = -\$5,140.82$ not justifiable

The sign changes between 15% and 20%:

$$\text{IRR} = 15 + 5 \times 282.33 / (282.33 + 5,140.82) = 15 + 0.26 = 15.26 \%$$

(The exact IRR is 15.24%. Interpolation lands slightly high because the NPV curve is convex. Write 15.26%.)
Accept if the required rate is below $\approx 15.2\%$.



The machine's NPV against the discount rate. IRR is where the curve crosses zero, and the chord gives the interpolated value.

4. Equipment with salvage [L18-19 · notes p10] [△ slide error](#)

Equipment costs 5,00,000. Life 4 years, generating an additional 1,60,000 of annual profit. "In the fifth year, the company plans to sell the equipment for its salvage value of 50,000." Calculate the IRR.

Q ▶ Which year does the 50,000 go in? Decide before reading on, because the two sources disagree.

Reading 1, the slide: the salvage arrives at the **end of year 4**, so the flows are $-5,00,000$, $1,60,000 \times 3$, then $2,10,000$.

NPV at 13% = $-5,00,000 + 1,60,000/1.13 + 1,60,000/1.13^2 + 1,60,000/1.13^3 + 2,10,000/1.13^4 = +6,581$
 NPV at 14% = ... same with 1.14 ... = $-4,202$
 IRR = $13 + 1 \times 6,581 / (6,581 + 4,202) = 13.61\%$

△ The slide prints $NPV_{13} \approx +9,635$ and $NPV_{14} \approx -6,774$. Both are wrong. The correct values are $+6,581$ and $-4,202$. Its final IRR $\approx 13.61\%$ is right (exact 13.61%).

Reading 2, the handwritten notes: the salvage arrives in **year 5** as a separate flow: $-5,00,000$, $1,60,000 \times 4$, then $50,000 \rightarrow$ IRR $\approx 13.28\%$ (notes: 13.27%).

In the exam: state which year you put the salvage in, then follow it through. The year-4 reading matches "life of four years" and is the professor's own slide answer.

Traps

- NPV must be **positive at i_1 and negative at i_2** . If both have the same sign, pick another rate.
- Interpolation is **linear**, so the answer is approximate. Say " $\approx$ ".
- IRR is compared against the **required rate**, not against 0.

Self-test

1. Cost ₹10,000, inflows ₹4,000/yr for 3 years. The NPV at 9% and 10% brackets zero. Find both NPVs and interpolate the IRR.
 2. Project NPV: +₹500 at 12%, –₹300 at 16%. Find the IRR.
 3. The IRR is 14% and the cost of capital is 11%. Decision?
-

Answers

1. 9%: $-10,000 + 4,000 \times 2.5313 = +\text{₹}125.2$. 10%: $-10,000 + 4,000 \times 2.4869 = -\text{₹}52.6$. IRR = $9 + 125.2/177.8 = \mathbf{9.70\%}$ (exact 9.70%).
2. $12 + 4 \times 500/800 = \mathbf{14.5\%}$
3. **Accept.** The project earns more than its capital costs.

10 — Mock MTE (30 marks, 90 min)

The format is assumed: there is no MEE2003 paper yet, so this copies the ECE2108 MTE sat the same week (A: 2+2+3 · B: 3 × 5 · C: 1 × 8). **New numbers throughout**, so this tests the method, not memory of the slide answers. Every answer was computed by script. Covers L1–L19 only (break-even and cost analysis haven't been uploaded).

Sit it closed-book, with a calculator, and write full working. **Answers are at the bottom.**

Section A (7 marks)

A1 (2) Differentiate between a *change in quantity demanded** and a change in demand*, with a sketch.

A2 (2) The price of a product falls from ₹40 to ₹36 and the quantity demanded rises from 500 to 600 units. Find the price elasticity of demand and classify it.

A3 (3) Annual demand 4,500 units. Ordering cost ₹100 per order. Unit price ₹50, carrying cost 20% of the unit price per year. Find the EOQ, the number of orders per year, and the cycle time in days.

Section B (15 marks)

B1 (5) Demand $Q_d = 120 - 4P$ and supply $Q_s = 20 + 6P$.

(i) Find the equilibrium price and quantity.

(ii) A tax of ₹5 per unit is imposed. Find the new equilibrium price and quantity, the Government's tax revenue, and the price the seller actually realises.

B2 (5) An investment of ₹1,00,000 yields ₹30,000, ₹40,000, ₹35,000 and ₹25,000 at the ends of years 1–4. The discount rate is 10%. Find (i) the payback period and (ii) the NPV. Should the project be accepted?

B3 (5) (a) Differentiate a ROM estimate from a definitive estimate (3).

(b) A past project of 12,000 lines of code took 1,800 person-hours. Estimate the effort for a similar 20,000-line project and name the technique (2).

Section C (8 marks)

C1 (8) A machine costs ₹80,000 and yields ₹25,000 a year for 5 years.

(a) Simple payback period and ROR (2)

(b) NPV at 10% (2)

(c) IRR by interpolation between 16% and 18% (3)

(d) Accept or reject if the required rate is 12%? Justify with both NPV and IRR (1)

Answers

A1 Change in **quantity demanded**: caused by the good's **own price** → **movement along** the curve (expansion or contraction). Change in **demand**: caused by **other factors** (income, tastes, related prices, population) → the **whole curve shifts** (right = increase, left = decrease). Sketch: one curve with two points on it, next to two parallel curves $D_1 \rightarrow D_2$.

A2 $\Delta Q = 100$, $\Delta P = -4$. $E_p = (100/-4) \times (40/500) = -2.0 \rightarrow |E_p| > 1$, **relatively elastic**.

A3 $H = 0.20 \times 50 = ₹10$.

$EOQ = \sqrt{(2 \times 4500 \times 100 / 10)} = \sqrt{90,000} = \mathbf{300 \text{ units}}$.

Orders = $4500/300 = \mathbf{15 \text{ per year}}$.

Cycle time = $300/4500 \text{ yr} = 0.0667 \text{ yr} = \mathbf{24.3 \text{ days}}$ (0.8 month).

(Inventory cost at the EOQ = $1,500 + 1,500 = ₹3,000/\text{yr}$.)

B1 (i) $120 - 4P = 20 + 6P \rightarrow \mathbf{P = ₹10, Q = 80}$.

(ii) $Q_s' = 20 + 6(P - 5) = 6P - 10 \rightarrow 120 - 4P = 6P - 10 \rightarrow \mathbf{P' = ₹13, Q' = 68}$.

Tax revenue = $5 \times 68 = \mathbf{₹340}$. The seller realises $13 - 5 = \mathbf{₹8}$. (Buyers carry ₹3 of the tax, sellers ₹2.)

B2 (i) Cumulative: $30,000 \rightarrow 70,000 \rightarrow 1,05,000$. It crosses in year 3: $2 + 30,000/35,000 = \mathbf{2.86 \text{ years}}$. (ii) PVs: $27,272.73 + 33,057.85 + 26,296.02 + 17,075.34 = 1,03,701.93$ NPV = $1,03,701.93 - 1,00,000 = \mathbf{₹3,701.93 > 0} \rightarrow \mathbf{accept}$.

B3 (a) ROM: earliest stage, minimal information, based on experience, accuracy -50% to +100%, used for go/no-go. Definitive: final planning stage, complete scope and specs, detailed cost breakdown, accuracy -5% to +10%, used for funding approval. (b) **Analogous** estimating: $1,800 \times 20,000/12,000 = \mathbf{3,000 \text{ person-hours}}$.

C1 (a) Payback = $80,000/25,000 = \mathbf{3.2 \text{ years}}$. ROR = $25,000/80,000 = \mathbf{31.25\%}$.

(b) $NPV_{10} = -80,000 + 25,000 \times 3.7908 = \mathbf{₹14,770 > 0}$.

(c) $NPV_{16} = -80,000 + 25,000 \times 3.2743 = \mathbf{+₹1,857}$. $NPV_{18} = -80,000 + 25,000 \times 3.1272 = \mathbf{-₹1,821}$. IRR = $16 + 2 \times 1,857/(1,857 + 1,821) = \approx \mathbf{17.01\%}$ (exact 16.99%). (d) **Accept**. NPV at 12% = $₹10,119 > 0$, and IRR $\approx 17\% > 12\%$.